# Channel-Aware Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

Peiyi Yue

Abstract—Connected and automated vehicles can improve perception reliability by exchanging information through vehicle-to-vehicle (V2V) links. However, cooperative perception must operate under limited bandwidth and time-varying vehicular channels. Existing communication-efficient perception methods mainly reduce redundancy within a fixed semantic representation, such as object-level detections or feature-level messages. This paper shows that fixed semantic granularity is suboptimal: feature-level messages are potentially more informative but can become unreliable or wasteful under poor channel quality, while object-level messages are compact and stable but less expressive. We propose Channel-Aware Task-Oriented Semantic Granularity Selection (CA-TOSG), a receiver-driven framework in which the ego vehicle selects, for each frame, one granularity from the deployed action set $\mathcal{S} = \{E, L, F\}$—ego-only, object-level, or feature-level (the denser 256-QAM feature mode is a physical-layer comparator, not a deployed action)—from its own perception cues and estimated channel state, and signals the choice to the collaborator via a 2-bit request piggy-backed on periodic V2X awareness messages, such as CAM/BSM, over 802.11bd or NR sidelink. CA-TOSG chooses one message type from object-level communication and compressed feature-level communication modes before any high-payload transmission. We derive the selector as the Lagrangian-relaxed solution of a constrained perception-communication problem and instantiate it with a 400-tree Random Forest. Experiments on the OPV2V dataset (validate, scene-disjoint test, and the Culver-City domain shift) under AWGN and Rayleigh channels report the realised true end-to-end AP@0.5 of the frozen selectors descriptively across three prescribed average communication budgets $B_{\max} \in \{0.10, 0.20, 0.30\}$. How much AP any granularity policy can contest on a split is bounded by that split's headroom, the gap between the perfect-channel feature-level ceiling and fixed object-level communication: 0.0550 on validate, 0.0240 on test and 0.0970 on Culver-City. Read the same way on every split, the frozen selector converts at most about a fifth of that headroom (up to 21.3%, and as little as 0.0% on Culver-City at the tightest budget), and we claim no AP advantage on any split. What the selector does buy is channel use: averaged over all channel states it spends 3.7–21.4% of the per-frame channel use of fixed feature-level transmission under rate-1/2 LDPC + 16-QAM coding, and under fading it falls back to the robust object-level message. The channel-type and estimated SNR features dominate the selector's feature importance, jointly accounting for 61.7% against 38.3% spread over 21 ego-side cues — importance, not sufficiency: with the channel features alone the selector stops requesting features at the tighter budgets. The result is therefore not an accuracy result: a one-line SNR threshold and a three-scalar hand rule reach comparable F1, but they do so by sending more feature messages, and what the task cues buy is knowing which of the frames the channel can carry are actually worth sending. The single pre-registered confirmatory comparison is against the nominal SNR-threshold rule on the scene-disjoint test split at $B_{\max} = 0.20$: the selector is non-inferior within the pre-registered 0.005 margin while reducing communication payload by 34.8%, a comparator that is itself over budget at $0.2168 > 0.20$ Msym. Against the budget-matched $\tau_{\text{feasible}}$ the selector is ahead on F1 by +0.00067 at 26.6% lower payload; that comparison is secondary and no non-inferiority decision is attached to it. Neither half of the input is sufficient on its own: given only the channel state the selector stops requesting features altogether at the two tighter budgets, and under deep fading the digital feature branch turns infeasible over the evaluated SNR range. Whether a graceful codec shifts that balance is exploratory evidence only, reported in an appendix under the prior protocol and not re-evaluated under the frozen one. The deployed selector incurs 52.1 ms per frame on a single CPU core (the slowest of the three frozen selectors), fitting the 100 ms budget of a 10 Hz LiDAR cycle. These results demonstrate that channel-aware semantic granularity adaptation is an effective strategy for bandwidth-constrained V2V cooperative perception.

Index Terms—Vehicle-to-vehicle communication, cooperative perception, connected vehicles, semantic communication, 3D object detection, channel-aware communication.

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## I. Introduction

Connected and automated vehicles are expected to rely not only on their onboard sensors, but also on information shared by surrounding vehicles. By exchanging perception information through vehicle-to-vehicle (V2V) links, a vehicle can obtain complementary viewpoints and improve its understanding of the driving scene. This capability is particularly important in safety-critical scenarios such as intersections, dense traffic, occluded road segments, and long-range perception, where a single vehicle's LiDAR or camera may provide incomplete observations. Cooperative perception therefore has the potential to improve detection robustness beyond the physical sensing range and viewpoint of an individual vehicle [1], [2].

Despite this potential, practical V2V cooperative perception is fundamentally constrained by communication. Sharing raw point clouds or dense intermediate feature maps can preserve rich geometric and semantic information, but the resulting payload can be too large for vehicular links with limited bandwidth and latency requirements. Moreover, V2V channels are time-varying due to mobility, blockage, multipath propagation, and fading [3], [4]. A communication strategy that is effective under a clean channel may become unreliable under low signal-to-noise ratio (SNR) or fading. Therefore, cooperative perception cannot be designed only as a perception problem; it must also account for the reliability and cost of the vehicular communication link.

Existing cooperative perception methods make different choices along the perception-communication tradeoff. Late-fusion methods transmit object-level predictions, such as 3D bounding boxes, class labels, and confidence scores [1]. These messages are compact and can be protected with strong coding, but they discard intermediate semantic context and may suffer when local detections are uncertain or incomplete. Intermediate-fusion methods transmit learned feature maps, which usually provide richer information for downstream fusion and detection [2], [5], but at a much higher communication cost. Recent communication-efficient methods reduce this cost by selecting important feature regions, applying attention-based masks, or compressing feature representations [5]. Semantic communication and joint source-channel coding methods further optimize the transmitted feature representation for task performance under wireless channels [6], [7].

However, most existing methods operate within a fixed semantic granularity. Object-level approaches always transmit object-level messages, while feature-level approaches always transmit feature-level messages, even if only a subset of feature tokens is selected. Thus, many methods answer the question of which feature elements should be transmitted, but do not explicitly answer a higher-level question: whether feature-level communication is needed for the current frame at all. This distinction is important in vehicular systems. In some frames, local perception may already be confident and stable, making compact object-level communication sufficient. In other frames, such as those with occlusion or ambiguous detections, richer feature-level communication may provide useful information for fusion. The best communication granularity is therefore not fixed, but depends on the task state of the perception system.

The communication channel further changes this decision. Feature-level messages are potentially more informative, but they are also more sensitive to channel degradation because larger and denser messages require more reliable transmission. Under poor SNR or fading, a feature-level message may be corrupted or effectively unusable, making its additional payload wasteful or even harmful for task performance. In contrast, object-level messages are much smaller and can be treated as robust low-rate information in many system designs. As a result, richer semantic communication does not always lead to better cooperative perception. The optimal message granularity depends jointly on perception task cues and channel quality.

Motivated by this observation, this paper proposes Channel-Aware Task-Oriented Semantic Granularity Selection (CA-TOSG) for bandwidth-constrained V2V cooperative perception. Instead of transmitting a fixed representation for every frame, CA-TOSG selects one message type from a set of semantic communication candidates. In the considered implementation, the selector chooses between ego-only operation, a compact object-level message and a compressed feature-level message, which differ substantially in per-frame channel use (a denser 256-QAM feature mode is defined but excluded as dominated; Section III-B). The decision is made using ego-side task cues, such as prediction confidence, visibility, temporal stability, and detection uncertainty, together with estimated channel quality such as SNR. The selected message is then transmitted through the V2V link and fused with the ego vehicle's local feature for 3D object detection. An overview of the resulting framework is shown in Fig. 1.

This formulation is complementary to existing feature compression and semantic communication methods. A feature codec, sparse feature selector [5], or joint source-channel coding module [6], [7] can be used inside the feature-level branch. CA-TOSG operates one level above these modules by deciding when feature-level communication should be activated. This is especially relevant for vehicular networks, where unnecessary high-payload communication should be avoided when the channel cannot support reliable feature transmission or when the current frame does not require additional semantic detail.

The main contributions of this paper are summarized as follows:

- We formulate channel-aware semantic granularity selection for V2V cooperative perception, where the ego receiver adaptively selects the semantic level it needs and requests it, according to its own task state and estimated channel quality; the collaborator transmits the requested level.
- We instantiate the framework as a lightweight, interpretable, receiver-driven policy that uses only ego-side perception cues and an estimated SNR/channel-type, adds no learnable parameters to the perception backbone, and whose inference fits within the 100 ms frame interval of a 10 Hz LiDAR cycle on a single CPU core (we measure the selector, not the end-to-end chain). The policy is deliberately model-agnostic: a 400-tree Random Forest [8] and a one-line SNR threshold are two policy realisations with different payload–F1 operating points, so the contribution is the selection framework, not a particular classifier.
- We provide a communication-aware evaluation protocol under AWGN and Rayleigh channels. The mainline comparison is between fixed ego-only, object-level and feature-level transmission with LDPC + 16/256-QAM coding [3], an SNR-threshold rule at both its nominal and its budget-matched tuning, two- and three-scalar hand rules over the same cues, and a channel-aware oracle. A learned semantic-communication baseline [6] is reproduced as an exploratory codec comparison under the prior protocol and is reported in Appendix A, not in the mainline tables.

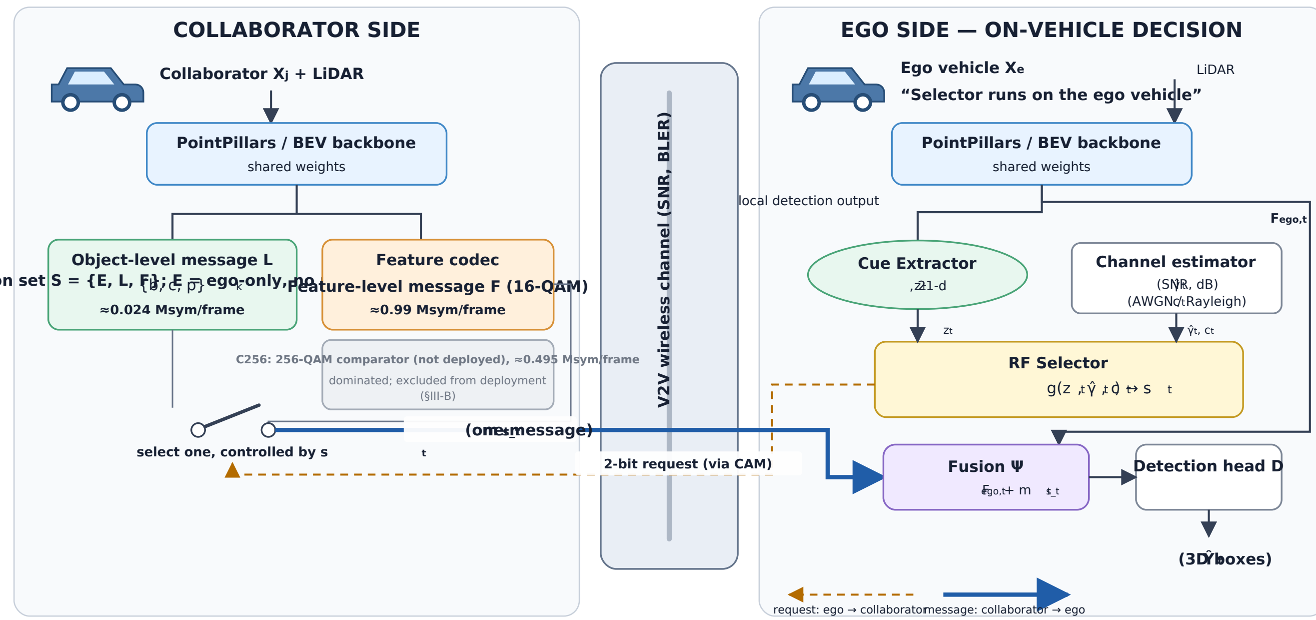


Fig. 1. Overview of the proposed CA-TOSG framework. The ego vehicle extracts online task cues $z_t$ and estimated channel quality $(\gamma_t, \mathrm{ch}_t)$, then selects one communication mode $s_t \in \mathcal{S} = \{E, L, F\}$, where $E$ is ego-only operation with no request at all (the dominated $C_{256}$ mode is a physical-layer comparator and is not in the deployed action set, Section III-B). Only the selected message is transmitted through the V2V link and fused with the local ego feature $F_{\mathrm{ego},t}$ for 3D object detection.

- Experiments on OPV2V show that CA-TOSG avoids unreliable high-payload feature transmission under poor channel conditions and approaches oracle performance when the channel supports richer communication. We characterise where it sits on the payload–F1 plane rather than claiming it is better on average, and report, per budget, the comparison against an SNR threshold tuned on validate for the same target budget, and show that, under a frozen selector, the communication saving transfers to the scene-disjoint test split and to the Culver-City domain shift (payload reductions of 32.2–49.2% and 42.1–68.9% against the nominal threshold, which is itself over budget at $B_{\max} = 0.20$ and 0.30), while the F1 non-inferiority does not extend to Culver-City at $B_{\max} = 0.20$, where the selector falls 0.00883 below the threshold rule (95% CI $[-0.00902, -0.00865]$), outside the pre-registered 0.005 margin.
- Under the evaluated LDPC + QAM transport, channel state carries most of the granularity decision — most, not all, and only on this transport: neither half of the input is sufficient on its own, and given only the channel state the selector stops requesting features at the two tighter budgets. Under deep fading the codec choice instead sets feasibility: the digital LDPC feature branch is infeasible over the evaluated SNR range, its frame-error threshold rising as the diversity order falls. Whether a graceful codec changes that currency is exploratory evidence only (Appendix A); it was not re-evaluated under the frozen protocol and no claim in this paper rests on it.

## II. Related Work

### A. V2V Cooperative Perception

V2V cooperative perception exploits information from surrounding connected vehicles to alleviate occlusion, long-range sparsity and limited field of view [1], [9]. Methods are categorised by the stage at which information is shared.

Early fusion shares raw point clouds before feature extraction [1]: the most complete information, but a very large payload requiring accurate spatio-temporal alignment, which is hard to deploy under practical V2V bandwidth and latency constraints.

Late fusion shares object-level predictions—boxes, classes, confidences [1]. The payload is small and suits low-rate links, but intermediate semantic and spatial information is discarded: an object missed by the sender cannot be recovered by the receiver, which costs accuracy in occluded or uncertain scenes.

Intermediate fusion shares learned BEV features between backbone and detection head [2], [5], [10]–[15], balancing the two extremes: richer spatial and semantic information than object-level sharing, but messages far larger, and expensive under bandwidth-constrained links.

Benchmarks such as OPV2V and OpenCOOD [1] provide common datasets and protocols for comparing these strategies. We build on that setting but ask a different question: rather than assuming a fixed fusion representation, how should the semantic granularity of the transmitted message be selected per frame from task and channel conditions?

### B. Communication-Efficient Cooperative Perception

To reduce the communication burden of intermediate fusion, many methods aim to transmit only task-relevant features. A common strategy is to use confidence maps, attention weights, or importance masks to select informative spatial regions from the feature map [5], [6]. For example, spatially selective methods transmit only a subset of BEV tokens that are expected to contribute to downstream detection. This reduces redundant communication while retaining much of the benefit of feature-level cooperation.

Where2comm [5] is a representative method in this direction. It estimates where communication is needed and selectively transmits spatial feature regions for cooperative perception. Such methods are effective because not all BEV locations contribute equally to object detection. Attention-based and confidence-based approaches follow a similar principle: they suppress unimportant feature regions and allocate communication resources to more informative areas.

Feature compression is another important direction. Instead of directly transmitting dense feature maps, the sender can encode them into a lower-dimensional representation before communication [6]. The receiver then decodes or fuses the compressed feature representation for detection. These approaches can reduce payload and may be combined with channel coding or learned semantic communication modules.

Most of these methods, however, optimise within a fixed semantic level: they assume feature-level transmission and choose which tokens or compressed representations to send. The closest exception is ML-Cooper [16], whose soft actor–critic policy adapts the proportion of raw-, feature- and object-level data to the network state—a heavyweight learned policy that does not characterise when granularity adaptation is required. Existing approaches either mix collaboration levels by channel condition [16], adapt content within the feature level [5], [6], or choose hybrid fusion by spatial overlap [17]. CA-TOSG instead makes a lightweight, interpretable, receiver-driven per-frame choice across object- and feature-level representations, from ego-side task cues and explicit channel reliability, under a practical digital PHY with delivery-failure fallback.

### C. Semantic and Channel-Aware Communication for Vehicular Perception

Semantic communication optimises the transmitted representation for downstream task performance rather than for reconstruction or bit error rate [6], [7], [18]–[20]; in cooperative perception it should preserve what detection needs, not the full feature map. A recent multi-mode IoT variant jointly selects the transmission mode and allocates resources across sensors [21]—a different axis from per-frame object-versus-feature granularity for V2V perception.

Joint source-channel coding and learned semantic codecs provide one way to improve robustness under noisy wireless channels [6], [7]. By training the encoder and decoder with channel effects in the loop, these methods can produce representations that degrade more gracefully than conventional separated source and channel coding under certain channel conditions. Digital communication baselines, such as modulation and coding schemes based on QAM and LDPC, often show threshold or waterfall behavior [3]: performance can be poor below a certain SNR and improve rapidly once the channel becomes reliable.

For vehicular perception, this channel dependence is especially important. V2V links experience time-varying SNR, multipath propagation, blockage, and fading [3]. A feature-level message that is useful under a clean channel may become unreliable under low SNR or Rayleigh fading. Therefore, the semantic value of a transmitted message depends not only on its content but also on whether the channel can support its reliable delivery.

Some make that transmission channel-adaptive at a fixed granularity: SmartCooper [22] adjusts the feature compression ratio from the estimated channel state, AccBEV [23] conditions a feature-repair module on the SNR. Both adapt how the feature message is encoded or repaired while still committing to feature-level transmission, so they are complementary: such a codec can sit inside CA-TOSG's feature branch, because CA-TOSG acts before the mode is chosen.

A concurrent line of work pushes semantic communication for collaborative perception into the digital domain. CoDS [24] pairs a task-oriented semantic compression codec with a semantic analog-to-digital converter, so that learned features traverse a standard digital V2X bitstream – LDPC-coded and QAM-modulated, the same cliff-prone class of digital transport we characterise – rather than an analog channel. On the sender side a feature-selection network identifies task-important spatial locations for transmission; on the receiver side an uncertainty-aware network discards features corrupted by decoding failures, mitigating the LDPC cliff after decoding. CoDS thus selects by task importance at the sender and filters by decode reliability at the receiver. CA-TOSG makes the prevailing channel state a first-class conditioning signal alongside the task: from local task cues and the estimated channel state it selects the message granularity per frame – ego-only operation with no request at all, the compact object-level message, or the feature-level message – and signals the choice through the 2-bit request before any high-payload transmission; when the channel cannot carry a feature message, the feature-level actions are gated out of the feasible set and the selector defaults to the compact object-level message. The two are complementary rather than competing – a digital semantic codec such as that of CoDS could serve as the feature-level branch that CA-TOSG activates, placing channel-conditioned granularity selection one level above codec and spatial-selection design. Both address the cliff effect, at different points: CoDS discards corrupted features after they are received, whereas CA-TOSG avoids spending the collaborator's transmission budget on an undeliverable feature message. This channel-aware semantic granularity selection is particularly suitable for vehicular systems, where communication resources are limited and channel quality can change rapidly.

## III. System Model and Problem Formulation

### A. Cooperative Perception Setting

We consider a single ego–collaborator V2V link. At each frame $t$, the ego vehicle observes the scene using its onboard sensors and extracts a local bird's-eye-view (BEV) feature representation, denoted by $F_{\text{ego},t}$. The collaborating vehicle also observes the scene from its own viewpoint and extracts a local feature representation $F_{j,t}$ using a shared PointPillars backbone [25]. The collaborator can transmit a perception message to the ego vehicle through the V2V link. The ego vehicle then fuses the received message with its local feature and produces the final 3D object detection output.

We adopt a receiver-driven signalling architecture: the ego vehicle evaluates its own perception state and channel state, decides which semantic granularity is needed, and signals the choice to the collaborator using a 2-bit request piggy-backed on the existing Cooperative Awareness Message (CAM), carried over IEEE 802.11bd or NR sidelink [3]. The request thus costs 2 bits per frame at 10 Hz, i.e., 20 bps, which is negligible relative to the 24 kbit per frame (0.24 Mbit/s at 10 Hz) $L$ payload and is already provisioned in the standard CAM signalling budget. The 2-bit field carries four codepoints; the deployed instance uses three—$E$ (ego-only, no request), $L$ and $F$—leaving the fourth codepoint and the non-deployed $C_{256}$ comparator unused. The collaborator complies by transmitting the requested message $m_t^{s_t}$. This receiver-driven design makes all perception cues used by the selector locally observable at the ego (Section IV-B) and avoids any causality issues that would arise if the collaborator had to estimate ego-side perception state without feedback.

Let $Y_t$ denote the ground-truth objects at frame $t$, and let $\hat{Y}_t$ denote the predicted detection output at the ego vehicle after fusion. The objective of cooperative perception is to improve the task utility of $\hat{Y}_t$, such as average precision (AP) or F1 score, while keeping the V2V communication payload low. Unlike conventional cooperative perception settings that assume a fixed type of transmitted representation, we consider an ego-side selector that can choose the semantic granularity of the requested message according to both perception-side task cues and the estimated channel condition.

### B. Message Candidates

At each frame, the ego receiver selects one communication mode from the deployed action set and requests it

$$s_t \in \mathcal{S} = \{E, L, F\}, \tag{1}$$

TABLE I
Notation for the semantic granularity levels. $F$ and $C_{16}$ denote the same message; the $C_q$ form is used only where the modulation order is the subject.

| Symbol | Meaning | Channel use | Role |
|---|---|---|---|
| $E$ | ego-only, no transmission | $B_E = 0$ | deployed action; also the deli |
| $L$ | object-level message | $B_L = 0.024$ | deployed action |
| $F \equiv C_{16}$ | feature-level, 16-QAM | $B_F = 0.99$ | deployed action |
| $C_{256}$ | feature-level, 256-QAM | 0.495 | physical-layer comparator, nc |

where $E$ denotes ego-only operation, $L$ object-level communication, and $F$ the compressed feature-level message, transmitted under 16-QAM modulation with rate-1/2 LDPC coding; we write $C_{16}$ for the same message where the modulation itself is the subject.

$E$ is a first-class action, not merely a failure mode. Choosing $E$ means the ego requests nothing and detects from its own LiDAR alone, so it costs $B_E = 0$ channel uses and cannot fail in transit. It is the right choice on two distinct kinds of frame: those where the ego's own detection is already sufficient, so any message would buy nothing, and those where the channel is so poor that a feature request would almost surely be lost and an object-level message would still cost bandwidth for little gain. $E$ also serves as the delivery-failure fallback: when a requested $F$ message does not arrive, the receiver falls back to the ego-only detection, which is why the effective utility of $F$ in Eq. (6) interpolates between the feature-level and the ego-only result according to the block-error probability. The two roles share one quantity but are decided at different times—$E$ as an action is chosen before transmission by the selector, whereas $E$ as a fallback is what remains after a delivery failure.

The same feature-level message is additionally evaluated with 256-QAM ($C_{256}$, same rate-1/2 LDPC) as a physical-layer comparator, but it is not included in the deployed semantic action set $\mathcal{S}$: modulation order is a transport parameter, not a semantic granularity (Section III-B). The transmitted message is therefore

$$m_t = m_t^{s_t}, \tag{2}$$

where only the message corresponding to the selected mode $s_t$ is transmitted. This one-of-many design is important: the sender does not transmit both object-level and feature-level messages simultaneously, but chooses the semantic granularity that is expected to provide the best task-communication trade-off for the current frame.

The object-level message is defined as a compact set of local detections:

$$m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}, \tag{3}$$

where $N_t$ is the number of detected objects at the sender, $b_k$ is the 3D bounding box, $c_k$ is the class label, and $p_k$ is the confidence score of object $k$.

The compressed feature-level message is defined as

$$m_t^{C_q} = \phi_q(F_{j,t}), \quad q \in \{16, 256\}, \tag{4}$$

where $\phi_q(\cdot)$ denotes a feature encoding and communication process for mode $C_q$. The two feature-level modes carry the same perception payload of approximately 1.98 Mbit but require different numbers of channel uses because of the different spectral efficiency of 16-QAM versus 256-QAM.

Of the two feature-level modes, the 256-QAM variant ($C_{256}$) carries the identical semantic content as $C_{16}$—the same compressed feature payload of Eq. (7)—and differs only in modulation order, which halves the per-frame channel use (0.495 vs. 0.99 Msym for $C_{256}$ and $C_{16}$[1]). It is included as a physical-layer comparator: holding the semantic content fixed and changing only the constellation is what isolates the codec's channel response in Fig. 2. It is not an element of the deployed action set $\mathcal{S} = \{E, L, F\}$ (Eq. (1)), and it is excluded by design rather than by measurement: the selector chooses what to send—a semantic granularity the receiver can request with the 2-bit field—whereas modulation order is a transport parameter of the link, not a granularity. Fig. 2 shows why the distinction is physically consequential: the 256-QAM frame-error cliff sits far to the right of the 16-QAM one (AWGN onset 16.5 vs. 8.0 dB[2]), and under Rayleigh and OFDM both flatline at the ego floor across 0–20 dB, so on this transport a denser constellation buys channel use at the cost of reach.

## C. Channel Model

We consider a V2V link whose quality varies across frames. Let $\gamma_t$ denote the estimated channel quality at frame $t$, represented by the SNR in dB. The channel quality may be obtained from pilot-based estimation, link-layer feedback, or reference-signal-received-power reports as already produced by IEEE 802.11bd and 5G NR sidelink stacks [3]. In this work, we use $\gamma_t$ as an explicit input to the semantic granularity selector in order to study how channel awareness affects cooperative perception.

Three representative channel models are considered. The first is the additive white Gaussian noise (AWGN) channel, which models noise-limited transmission and provides a controlled setting for studying SNR-dependent behavior. The second is the Rayleigh fading channel, which captures multipath fading. The third is a frequency-selective OFDM link over a tapped-delay-line channel with an exponential power-delay profile at $\tau_{\mathrm{rms}} = 46.2$ ns (3GPP TR 37.885 Urban NLOSv [4]), whose cross-subcarrier LDPC coding carries frequency diversity; a log-log slope fit of its codeword BLER over the waterfall region gives an empirical diversity order of 2.01, versus ≈1 for flat Rayleigh and the sharp cliff of AWGN. All per-frame block-error rates are tabulated from a Sionna link-level simulation. The feature-level message under mode $C_q$ traverses the channel as a coded LDPC block, with block-error rate (BLER) function $\mathrm{BLER}_q(\gamma)$ that is monotonically non-increasing in $\gamma$ and exhibits the threshold behavior characteristic of LDPC decoding. This is an all-or-nothing full-frame delivery model: a frame's feature message is usable only if all $N_{\mathrm{cw}}$ of its constituent codewords decode, with no codeword-level retransmission, erasure protection, or HARQ. We bound what that assumption costs rather than only naming it: splitting the frame into $k$ independently coded fragments that are all required leaves the frame BLER algebraically unchanged, and adding one retransmission per fragment lowers the marginal AWGN cliff point at 8 dB from 0.402400 to 0.100363 ($k = 2$) and 0.057077 ($k = 4$) at 1.226954× and 1.120770× the payload — but does not open the Rayleigh branch at all, whose frame BLER stays at 1.0 across the whole evaluated 0–20 dB range for every $k$ at 2.0× the payload. The Rayleigh infeasibility reported here is therefore a property of the link budget rather than of the no-HARQ assumption, while the AWGN thresholds would indeed move under partial recovery. Carrying the modified block-error rates and the inflated payload $B_F\,(1 + q_{\mathrm{first}})$ through the full deployment replay leaves every conclusion in place: over all three splits, all three budgets and $k \in \{1, 2, 4\}$, one retransmission moves the selector by at most +0.00064 F1 and +0.00803 Msym, and the SNR-threshold rule not at all at $B_{\max} = 0.20$, because its feature requests already sit above the cliff (results/sensitivity/r25_fragmentation_replay.csv). We treat the object-level message $L$ as a low-rate robust message in the evaluated SNR range, motivated by its small payload (approximately 0.024 Mbit) that allows aggressive channel coding or repeated transmission. Fig. 2 plots the resulting $\mathrm{BLER}_q(\gamma)$ for the LDPC + 16/256-QAM configurations under both channels, making explicit the threshold (cliff) behaviour that the selector must track.

## D. Effective Task Utility under Channel Corruption

Let $F_t^s$ denote the realized frame F1 of the ego vehicle's detection output when the ego receiver selects and requests mode $s$ at frame $t$. We model the effective frame F1 under channel corruption as

$$F_t^L = F_t^{L,\mathrm{clean}}, \tag{5}$$

$$F_t^{C_q} = (1 - \mathrm{BLER}_q(\gamma_t, \mathrm{ch}_t))\, F_t^{C_q,\mathrm{clean}} + \mathrm{BLER}_q(\gamma_t, \mathrm{ch}_t)\, F_t^{\mathrm{ego}}, \quad q \in \{16, 256\}, \tag{6}$$

where $\mathrm{ch}_t \in \{\mathrm{AWGN}, \mathrm{Rayleigh}\}$, $F_t^{\cdot,\mathrm{clean}}$ denotes the frame F1 under a perfect channel, and $F_t^{\mathrm{ego}}$ is the frame F1 of the ego vehicle's own single-agent (pre-fusion) detection. Equation (6) is an ego-fallback block-loss model: with probability $\mathrm{BLER}_q$ the feature-level message is lost and the receiver reverts to its ego-only output $F_t^{\mathrm{ego}}$; with probability $1-\mathrm{BLER}_q$ it is delivered intact and contributes its clean F1. Because only one message is requested per frame, a lost feature block leaves no collaborator information, so the fallback is the ego-only detection rather than a zero-utility state; both this frame-F1 model

[1] Feature payload 1.98 Mbit (Eq. (7)) at LDPC rate-1/2 gives 3.96 Mbit of coded bits; ÷8 bit/sym (256-QAM) = 0.495 Msym, ÷4 bit/sym (16-QAM) = 0.99 Msym.

[2] Es/N0 onset at which the frame-level BLER first falls below 0.999 – the same 0.999 constant that defines the feasibility mask. Fig. 2 plots both cliffs directly.

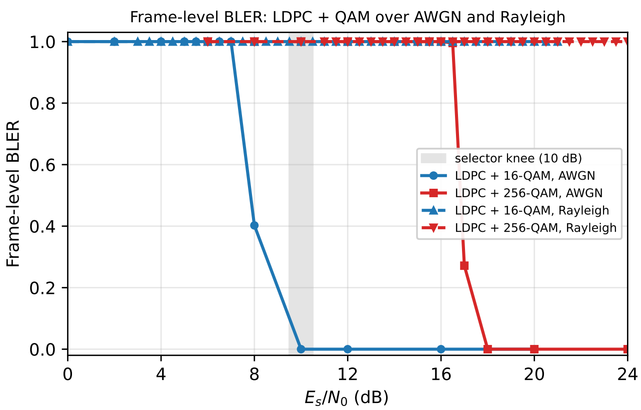


Fig. 2. Block-error rate of the feature-level message under LDPC 1/2 with 16-QAM and 256-QAM, over AWGN (solid) and Rayleigh (dashed) channels. The 16-QAM AWGN cliff (an 8–12 dB transition, onset 8.0 dB) defines the selector knee (shaded); the 256-QAM cliff sits far to its right (onset 16.5 dB), so by the time 256-QAM is reliable the 16-QAM block is already error-free. Under Rayleigh fading both curves degrade so slowly that, under the all-or-nothing frame model evaluated here, feature-level transmission never becomes reliable in 0–20 dB, explaining the selector's conservative $L$-only behaviour.

and the true end-to-end AP evaluation of Section VI-F realise the same per-frame fallback.

## E. Communication Cost

Let $B_s$ denote the per-frame payload in channel-use-equivalent mega-symbols (Msym) for mode $s$, accounting for the spectral efficiency of the modulation:

$$B_L = 0.024, \quad B_F = 3.96/4 \approx 0.99, \quad B_{C_{256}} = 3.96/8 \approx 0.495 \tag{7}$$

all in Msym/frame, where $3.96 = 1.98/(1/2)$ is the coded-bit count produced by rate-1/2 LDPC from the 1.98-Mbit feature payload, and the divisors 4 and 8 are the bits-per-symbol of 16-QAM and 256-QAM, respectively. The object-level message is sent as a protected low-rate message at approximately 1 bit per channel use (rate-1/2 QPSK), so its 0.024 Mbit of information occupies $\approx$0.024 Msym. The average communication cost over $T$ evaluated frames is

$$\bar{B} = \frac{1}{T}\sum_{t=1}^{T} B_{s_t} = \sum_{s\in\mathcal{S}} \rho_s B_s, \tag{8}$$

where $\rho_s$ is the fraction of frames assigned to mode $s$. This equation highlights the role of semantic granularity selection: reducing the frequency of high-payload feature-level transmission directly reduces the average communication cost.

## F. Task-Communication Objective

The goal of channel-aware semantic granularity selection is to choose a policy that balances detection utility and communication cost. Let $z_t$ denote the online task cue vector extracted from the ego-side perception stream, and let $\gamma_t$ denote the estimated channel quality. The selector is defined as

$$s_t = g(z_t, \gamma_t, \mathrm{ch}_t), \tag{9}$$

where $g(\cdot)$ maps the current task state and channel state to one of the message modes in $\mathcal{S}$.

In this paper we instantiate the policy via a channel-aware utility argmax,

$$s_t^\star = \arg\max_{s\in\mathcal{S}} F_t^s, \tag{10}$$

which is the $\lambda = 0$ special case of the payload-penalised objective introduced in Eq. (13) and adopted as the formal training target. Of the three deployed budgets only $B_{\max} = 0.30$ freezes at $\lambda^\star = 0$ and is therefore described by this form; the other two carry a strictly positive penalty.

## G. Derivation: Constrained Optimisation and Rate-Distortion View

Equation (10) is the Lagrangian-relaxed optimum of the constrained perception-communication problem

$$\max_{g} \quad \mathbb{E}[F_t^{s_t}] \quad \text{s.t.} \quad \mathbb{E}[B_{s_t}] \le \bar{B}_{\max}, \tag{11}$$

where $\bar{B}_{\max}$ is the prespecified average communication budget—a bound on the mean per-frame payload over a split, not a per-frame cap. Introducing the Lagrange multiplier $\lambda \ge 0$, the relaxed problem is

$$\max_{g} \; \mathbb{E}[F_t^{s_t}] - \lambda\,\mathbb{E}[B_{s_t}]\,, \tag{12}$$

whose pointwise optimum is

$$s_t^\star(\lambda) = \arg\max_{s\in\mathcal{S}} \big(F_t^s - \lambda B_s\big). \tag{13}$$

The learned selector is trained against Eq. (13), not against Eq. (10): the payload penalty $\lambda$ is a selected quantity, frozen per budget by the walk of Section V at $\lambda^\star = 0.05$, 0.02 and 0.00 for $B_{\max} = 0.10$, 0.20 and 0.30 respectively. Equation (10) is thus the $\lambda = 0$ member of the family and describes only the $B_{\max} = 0.30$ operating point; the two tighter budgets are trained with a strictly positive penalty, and the constrained problem itself is addressed by sweeping $\lambda$ over the family (Section VI-H).

From a rate-distortion perspective, the effective F1 model of Eq. (6) is a channel-induced distortion that scales with $\mathrm{BLER}_q(\gamma)$. The compressed-feature mode $C_q$ delivers "rate" $R_{C_q} = 3.96/\log_2 M_q$ in channel-use megasymbols (Msym)—0.99 and 0.495 Msym for $C_{16}$ and $C_{256}$—modulated by a channel-induced packet survival probability $1 - \mathrm{BLER}_q(\gamma)$, while the object-level mode $L$ delivers a much lower rate but with effectively unit survival probability under the protected low-rate transmission assumed in this paper. The selector therefore implicitly trades off rate against rate-conditional distortion as a function of the channel state, a structure familiar from joint source-channel coding theory but here expressed at the granularity-action level rather than at the symbol level.

This formulation leads to a key design principle: the most informative message is not always the most useful message under communication constraints. A feature-level message may provide richer semantics, but if the channel is poor, its realized task utility may be lower than that of a compact object-level message. Therefore, the selector must consider both the task state and the channel state before deciding the semantic granularity of the transmitted message.

## IV. Proposed Method

### A. Overview

The proposed CA-TOSG framework is designed to select the semantic granularity of V2V communication according to both the perception task state and the channel condition. Instead of transmitting a fixed representation for all frames, the ego vehicle chooses one message mode from $\mathcal{S} = \{E, L, F\}$—ego-only, object-level, or feature-level—before any high-payload transmission, signals the choice to the collaborator via the 2-bit CAM-embedded request introduced in Section III, and the collaborator complies.

Fig. 1 illustrates the overall pipeline. The ego vehicle first extracts a local BEV feature and local perception outputs using a shared PointPillars backbone [25]. Based on these outputs, it constructs an online task cue vector $z_t$ (Section IV-B). Meanwhile, the current channel quality is represented by the estimated SNR $\hat{\gamma}_t$ and the channel type $c_t$, both produced by the ego's receiver-side link adaptation module already present in 802.11bd / 5G NR sidelink stacks. A lightweight selector takes $(z_t, \hat{\gamma}_t, c_t)$ as input and predicts the communication mode $s_t$. The collaborator transmits the requested message: if $s_t = L$, compact object-level detections; if $s_t = F$, a compressed feature-level message under 16-QAM; if $s_t = E$ the ego issues no request at all and operates on its own perception. At the ego, the received message is fused with $F_{\text{ego},t}$ to produce the final detection result.

The key idea is that semantic richness and task utility are not equivalent under channel constraints. Feature-level messages contain richer intermediate representations, but they are more sensitive to channel degradation and require higher communication resources. Object-level messages are less expressive but compact and robust. Therefore, the selector should activate feature-level communication only when the channel condition and task state indicate that the additional semantic information is likely to improve detection.

### B. Ego-Side Online Task Cues

The first input to the selector is the ego-side task cue vector $z_t \in \mathbb{R}^{21}$, summarized in Table II. Every cue is computed from the ego vehicle's own LiDAR observation and the ego's own local detection stream; no information from the collaborator's perception stack is required at decision time. This is a direct consequence of the receiver-driven signalling architecture introduced in Section III and removes the causality issue that would arise if the collaborator had to estimate ego-side state without feedback. Pre-transmission, the ego runs the selector, signals the chosen mode $s_t$ to the collaborator via the 2-bit CAM request, and only then does the collaborator transmit $m_t^{s_t}$.

TABLE II
Ego-side online task cues used by the CA-TOSG selector ($d = 21$). All cues are computed locally on the ego vehicle before message selection, removing the causality issue that would arise from depending on the collaborator's state.

| Category | Cue | Definition |
|---|---|---|
| Agent context | num_cavs<br>ego_num_objects | number of CAVs in scene<br>ego-detected objects |
| Frame shape | ego_origin_lidar_shape_0/1 | raw LiDAR tensor shape |
| Range stats | pcd_num_points<br>pcd_mean_range<br>pcd_max_range<br>pcd_std_range | total point count<br>mean LiDAR range<br>max LiDAR range<br>range std. dev. |
| Distance bins | pcd_near_20m<br>pcd_mid_20_50m<br>pcd_far_50_80m<br>pcd_very_far_80m | points within 20 m<br>points 20–50 m<br>points 50–80 m<br>points beyond 80 m |
| Quadrant cnt. | pcd_front_points<br>pcd_back_points<br>pcd_left_points<br>pcd_right_points | front-half points<br>rear-half points<br>left-half points<br>right-half points |
| Long-range | pcd_front_far_30m<br>pcd_front_far_50m | front points $> 30$ m<br>front points $> 50$ m |
| Density bins | pcd_density_0_20<br>pcd_density_20_50<br>pcd_density_50_80 | density 0–20 m<br>density 20–50 m<br>density 50–80 m |

These cues collectively describe local sensing quality, scene geometry, and detection load, all of which influence whether the current frame benefits from richer feature-level information from a collaborator. Confidence-related and temporal-stability cues can be added straightforwardly when intermediate detection outputs are accessible to the cue extractor; the framework does not depend on any particular cue choice.

### C. Channel Quality Cue

The second input is the channel quality, represented by the estimated SNR $\hat{\gamma}_t \in \mathbb{R}$ in dB and a binary channel-type indicator $c_t \in \{0, 1\}$ encoding AWGN versus Rayleigh fading. In a deployed V2X stack, $\hat{\gamma}_t$ is obtained from pilot-symbol SNR estimation already produced by 802.11bd or 5G NR sidelink receivers [3], and $c_t$ can be inferred from link-layer multipath indicators such as RMS delay spread or Doppler estimates.

The role of $(\hat{\gamma}_t, c_t)$ is fundamentally different from the perception cues. Task cues estimate whether the current frame needs richer semantic information; the channel cues estimate whether such information can be transmitted reliably. A difficult frame does not necessarily imply that feature-level communication should be used; if the channel is poor, transmitting a corrupted feature-level message may not improve detection and may waste bandwidth. Conversely, when the channel is strong, feature-level communication may become worthwhile for frames where object-level messages are insufficient. The selector must therefore consider the joint state $(z_t, \hat{\gamma}_t, c_t)$, which is the foundation of CA-TOSG.

### D. Message Branches

1) Object-Level Branch: The object-level branch transmits compact detection results $m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}$. Its average payload is approximately $B_L = 0.024$ Msym/frame, allowing it to be transmitted with strong channel coding and treated as channel-invariant in the evaluated SNR range.

2) Feature-Level Branches: The feature-level branches transmit compressed BEV representations $m_t^{C_q} = \phi_q(F_{j,t})$, where $\phi_q(\cdot)$ uses $q$-QAM modulation under rate-1/2 LDPC coding. The $C_{16}$ mode is more reliable but less spectrally efficient (channel-use payload $B_{C_{16}} \approx 0.99$ Msym/frame), while the $C_{256}$ mode is more spectrally efficient but more channel-sensitive (channel-use payload $B_{C_{256}} \approx 0.495$ Msym/frame).

### E. Channel-Aware Semantic Granularity Selector

The selector implements the policy $g(z_t, \hat{\gamma}_t, c_t)$ of Eq. (9) as a Random Forest classifier. The three deployed selectors are the ones the pre-registered budget walk froze (FROZEN_MANIFEST.json): $N_T = 400$ trees and max_features=sqrt throughout, minimum samples per leaf of 2, a depth bound of 10 at $B_{\max} = 0.10$ and 0.20 and unbounded depth at $B_{\max} = 0.30$, and class_weight=None at all three budgets — the walk selected unweighted candidates, so no class re-weighting is applied to the oracle-label imbalance. We choose Random Forest for three practical reasons. First, the selector should be training-free with respect to the perception backbone and detection head: it should be deployable on top of any pretrained cooperative perception stack without retraining the perception model. Random Forest fits this constraint because it operates purely on tabular cues. Second, Random Forest provides interpretable feature importance (Section VI-D), which is valuable for safety certification of vehicular systems. Third, its per-frame inference cost on a single CPU core is $52.1 \pm 5.6$ ms (P95 = 58.3 ms), measured over 1,000 batch-1 trials per frozen selector and quoted for the slowest of the three (Section VI-O); this fits the 100 ms budget of a 10 Hz LiDAR cycle without requiring a GPU, dedicated accelerator, or framework-level optimisation. We emphasise that the contribution is not the Random Forest itself: it is an interpretable implementation of the channel-aware granularity policy. The SNR-threshold and three-scalar hand-rule baselines achieve comparable F1, but require higher communication payload at the primary operating point (Section VI-K).

The selector is trained to imitate the channel-aware oracle defined in Eq. (10). The oracle uses ground-truth detection outputs and is therefore not deployable; it is used only to generate training labels and as an upper-bound reference. Before taking the argmax in Eq. (10), the oracle applies a feasibility mask: any mode whose frame-level block-error rate exceeds 0.999 at the operating $(\gamma_t, \mathrm{ch}_t)$ grid point is removed from $\mathcal{S}$, so that neither the oracle labels nor the deployed selector request a message the channel almost surely cannot deliver; when a requested feature block is nonetheless lost, the pipeline reverts to the ego-only output rather than a zero-utility state (Eq. (6)). At deployment, the selector uses only the online cues $z_t$ and the channel state $(\hat{\gamma}_t, c_t)$:

$$s_t = g(z_t, \hat{\gamma}_t, c_t). \tag{14}$$

This design gives the selector an interpretable behavior. At low SNR, feature-level messages are likely to be unreliable, so the selector tends to choose $L$. At moderate SNR, $C_{16}$ may become useful because it provides richer features with better reliability than $C_{256}$. Under fading channels, the selector remains more conservative because the average SNR does not fully eliminate deep fading risk. We will see in Section V that the empirical behavior of the trained selector follows this pattern.

### F. Receiver-Side Fusion and Detection

After transmission, the ego vehicle receives the selected message $m_t^{s_t}$. The receiver fuses this message with the local ego feature:

$$\hat{Y}_t = D\left(\Psi(F_{\mathrm{ego},t}, m_t^{s_t})\right), \tag{15}$$

where $\Psi(\cdot)$ denotes the cooperative fusion module and $D(\cdot)$ is the detection head. The fusion operation depends on the selected message type: if $s_t = L$, the receiver performs object-level fusion using the received detections; if $s_t = F$, the receiver decodes the feature-level representation and performs intermediate fusion.

### G. Discussion of Design Advantages

The proposed design has three main advantages. First, it avoids unnecessary feature-level communication in easy frames or poor channels, reducing the average payload. Second, it prevents unreliable feature messages from degrading perception under low SNR or fading. Third, it remains compatible with existing feature-level communication modules. For example, Where2comm [5], JSCC-based codecs [6], or other semantic communication methods can be used inside the $C_{16}$ or $C_{256}$ branches, while CA-TOSG decides when these branches should be activated. This separation between semantic granularity selection and feature-level transmission is important for practical vehicular systems. It allows the communication policy to adapt at the frame level without redesigning the entire perception backbone or communication codec.

## V. Experimental Setup

### A. Dataset and Implementation

We evaluate CA-TOSG on the OPV2V dataset [1] using the OpenCOOD codebase. OPV2V is a large-scale simulated V2V cooperative perception dataset that provides multi-agent LiDAR observations, vehicle poses, and 3D object annotations under connected-vehicle settings. We use the validate split, which contains $T = 1{,}980$ frames, for all primary results, and additionally verify generalisation on the held-out OPV2V test split ($T = 2{,}170$ frames, scene-disjoint from validate) and the Culver-City split ($T = 550$ frames, a real-road-layout domain shift) in Section VI-G, applying the validate-trained selector without retraining. The perception model is a PointPillars BEV backbone [25] with detection range $x \in [-140.8, 140.8]$ m and $y \in [-38.4, 38.4]$ m. The backbone and detection head are taken from the public OpenCOOD pretrained checkpoints and are frozen throughout this work; CA-TOSG adds no learnable parameters to the perception pipeline. All methods share the same backbone and detection head to ensure a fair comparison.

## B. Message Construction and Payload Accounting

Both payloads are made explicit rather than left implicit: the object-level payload is derived from first principles, while the feature-level source budget is a declared convention to which, as shown below, all conclusions are insensitive. The object-level message $L$ carries the collaborator's detected objects; on the OPV2V validate split a frame contains on average 27 detected objects, and encoding each as an ETSI-CPM-style perceived-object container (3D box, dimensions, class, confidence and position covariance, ≈110 bytes) gives $B_L \approx 27 \times 110 \times 8 \approx 0.024$ Mbit/frame—a deliberately conservative figure that, if anything, overstates the object-level cost. The feature-level message encodes the transmitted BEV feature tensor of size $256 \times 48 \times 176 \approx 2.16 \times 10^6$ elements: the $281.6 \times 76.8$ m detection range at 0.4 m voxels yields a $704 \times 192$ pillar grid, down-sampled by 4 to $48 \times 176$ with 256 channels. We adopt a fixed source budget of $B_C \approx 1.98$ Mbit/frame for the feature message, i.e. ≈0.92 bit per element of this $2.16 \times 10^6$-element tensor; because every comparison in this paper is ratio-like (channel-use fractions and payload–F1 orderings), the conclusions are insensitive to this constant—re-anchoring to 2.16 Mbit would rescale the feature cost of all policies equally. The feature-level message is therefore ≈82× the object-level payload. Both feature modes $C_{16}$ and $C_{256}$ carry this same 1.98 Mbit perception payload but require different numbers of channel uses; applying the rate-1/2 coded-bit conversion of Eq. (7) gives $B_{C_{16}} \approx 0.99$ Msym/frame for 16-QAM and $B_{C_{256}} \approx 0.495$ Msym/frame for 256-QAM. All payload comparisons in this paper use this channel-use-equivalent definition.

## C. Channel Settings

We sweep the estimated SNR over $\gamma \in \{0, 2, 4, \ldots, 20\}$ dB, giving 11 SNR points per channel. Training and the headline evaluation use two channel models, AWGN and Rayleigh fading; the frequency-selective OFDM link of Section III-C (TDL, $\tau_{\mathrm{rms}} = 46.2$ ns, empirical diversity order 2.01) is additionally used in the codec-comparison analysis of Appendix A. The selector's training substrate is the full deterministic grid—every one of the 1,980 frames crossed with all 11 SNR points and both channel types (43,560 rows)—rather than one sampled $(\gamma, \mathrm{ch})$ per frame; the 200-realisation Monte-Carlo draw is used for evaluation, not for building labels. The BLER functions $\mathrm{BLER}_q(\gamma, \mathrm{ch})$ are tabulated from a separate LDPC + QAM simulation; we instantiate this transport with the 5G NR LDPC (Sionna) under 3GPP TR 37.885 Urban NLOSv [4]. This transport configuration—rate-1/2 LDPC with 16-/256-QAM, compared at the same coded-bit count—matches the conventional digital baseline of SComCP [7]. Rayleigh is simulated directly at link level in the same Sionna chain as AWGN—per-codeword flat block fading with $|h|^2 \sim \mathrm{Exp}(1)$ and perfect receiver CSI, so each block sees its own effective $E_s/N_0$—and is not obtained by averaging the AWGN table over an instantaneous-SNR distribution. Frame-level BLER follows from the measured codeword BLER as $1 - (1 - \mathrm{BLER}_{\mathrm{cw}})^{N_{\mathrm{cw}}}$ with $N_{\mathrm{cw}} = 3{,}960$ codewords per feature frame.

## D. Selector Training

The selector is a Random Forest classifier [8] over the 23-dimensional input $x_t = [z_t, \hat{\gamma}_t, c_t]$, with labels the channel-aware oracle $s_t^\star$ of Eq. (10). Selection is budget-indexed and pre-registered rather than hand-tuned: a fixed block of 112 candidates (hyperparameter settings × a payload-penalty grid $\lambda$) is evaluated once under scene-level 9-fold leave-one-scene-out cross-validation on validate—each of the nine scenes held out exactly once, 1,008 fold rows—and for each budget $B_{\max} \in \{0.10, 0.20, 0.30\}$ the candidate list is walked in descending out-of-fold F1 until one meets the budget, which freezes that budget's $(\lambda^\star, \tau^\star)$ and hyperparameters. The three frozen selectors are $\lambda^\star = 0.05/0.02/0.00$ with $\tau^\star = 18/12/8$ dB at $B_{\max} = 0.10/0.20/0.30$; all three carry class_weight=None—the balanced candidates were walked past for exceeding their budget—and FROZEN_MANIFEST.json is the authority for every one of these values. Each frozen configuration is then refitted once on the full 1,980-frame validate split and frozen—never retrained—for every reported test and Culver-City result. Run-to-run variance is exposed not by a within-validate resplit but by averaging all realised-F1 and payload metrics over 200 Monte-Carlo channel realisations per operating point (per-frame SNR $\sim \mathcal{U}[0, 20]$ dB and channel type $\sim$ Bernoulli(0.5) AWGN/Rayleigh).

## E. Baselines

We compare CA-TOSG with the following baselines:

Fixed $L$. The sender always transmits the object-level message. This is treated as channel-invariant in the evaluated SNR range.

Fixed $F$. The sender always transmits the feature-level message under 16-QAM coding.

Fixed $C_{256}$. The sender always transmits the feature-level message under 256-QAM coding.

SNR-threshold rule ($\tau^\star$). A one-line channel rule—request $F$ under AWGN when $\hat{\gamma}_t > \tau^\star$, otherwise $L$—with $\tau^\star$ tuned on validate for each budget. Reported both at its nominal tuning and at the strictly budget-matched $\tau_{\text{feasible}}$.

Two- and three-scalar hand rules. Threshold policies over the same committed cues, fitted through the identical scene-level LOSO walk and budget constraint as the selector: a two-scalar rule (difficulty gate plus link-reliability gate) and, after the two-scalar form proved unable to meet the two tighter budgets, a three-scalar amendment whose feature gate is a conjunction of the two. They are the strongest hand-designed comparators in this paper and are analysed in Section VI-K.

Channel-aware oracle. For each frame, the oracle selects the message mode that maximises Eq. (6) given the realised SNR and channel; it is not deployable but serves as an upper-bound reference.

ImportanceMapJSCC (exploratory; Appendix A). The learned semantic communication baseline of [6], evaluated under the same perception backbone and the same AWGN/Rayleigh channel models. It is a prior-protocol arm: it was not re-evaluated under the frozen single-collaborator protocol and therefore appears in no mainline table or figure—every result involving it is confined to Appendix A and is exploratory. This is the strong feature-level communication baseline. It is our reproduction of the published method under our perception backbone, dataset split and channel tables—no code, checkpoint or trained selector from the original authors is used—so it should be read as a faithful re-implementation rather than as the authors' own system.

LDPC + 16/256-QAM (separate coding). Standard separate source-channel coding baselines using rate-1/2 LDPC with 16-QAM and 256-QAM modulation [3]. These represent traditional digital communication of the same feature-level payload as ImportanceMapJSCC.

Perfect-channel upper bound. Detection performance under ideal (lossless) feature-level transmission, included as a saturating ceiling for the AP curves.

### F. Evaluation Metrics

We report AP@0.5 and AP@0.7 for end-to-end detection performance, mean realised frame F1 for fine-grained frame-level behaviour, and average payload in channel-use-equivalent Msym/frame for communication cost. We also report the selection ratios $\rho_s = (1/T)\sum_t \mathbf{1}(s_t = s)$ to expose how the selector's policy varies with channel quality.

## VI. Results and Analysis

### A. Headline Results

Because the value of feature-level cooperation is realised only when the channel can deliver it, a single channel-averaged number understates the effect: under Rayleigh fading and low-SNR AWGN the selector deliberately holds the object-level message $L$, so averaging over all channel states dilutes the gain. We therefore report the true end-to-end detection AP in the two regimes that define the policy: the fallback regime (fading or low-SNR AWGN), where CA-TOSG holds $L$ at 0.024 Msym/frame, and the feature-active regime (AWGN $\hat{\gamma}_t \geq 10$ dB, the measured $\rho_F$ knee of Section VI-F rather than a stipulated threshold), where the selector requests $F$. AP is computed by concatenating cached per-frame predictions according to the selector's choice and scoring against OPV2V ground truth (full protocol in Section VI-F); the per-SNR breakdown and the Rayleigh curves are in Tables VII–X.

Three observations follow. (i) The realised true end-to-end AP@0.5 of the frozen selectors, reported descriptively for $B_{\max} = 0.10/0.20/0.30$, is 0.7887/0.7926/0.7936 on validate, 0.8697/0.8742/0.8742 on test and 0.7299/0.7347/0.7504 on Culver-City, against fixed object-level references of 0.7819, 0.8691 and 0.7299. Every split is read the same way: its AP headroom is the gap between the perfect-channel feature ceiling and fixed object-level communication — 0.0550 on validate, 0.0240 on test and 0.0970 on Culver-City — and the selector's realised share of that headroom is $(\text{AP}_{CA-TOSG} - \text{AP}_{\text{Fixed } L})$/headroom: 12.4/19.5/21.3% on validate, 2.5/21.2/21.2% on test and 0.0/4.9/21.1% on Culver-City. The headroom exists on all three splits, but the selector converts at most about a fifth of it, so what it reliably delivers is the communication saving rather than an AP gain. Under the single-collaborator convention the $L$–$F$ gap is wider than it was — box-level fusion of one neighbour discards more than feature-level fusion of the same neighbour — which is why the retired full-collaborator accounting showed a test headroom roughly an order of magnitude smaller: that near-zero was an artefact of fusing every collaborator into both branches. These figures are descriptive and we draw no AP advantage from them. The quantity that bounds what any granularity policy could contest is the per-split headroom between the perfect-channel feature-level ceiling and fixed $L$: $0.8369 - 0.7819 = 0.0550$ AP on validate, $0.8931 - 0.8691 = 0.0240$ AP on the scene-disjoint test split and $0.8269 - 0.7299 = 0.0970$ AP on the Culver-City domain shift. Test has the smallest headroom of the three (0.0240 AP) but it is no longer negligible, and the frozen selector converts about a fifth of it; the split is therefore informative about payload and about not losing accuracy, not about AP gain. (ii) Under fading or low SNR the selector holds $L$ and incurs no AP loss relative to object-level communication, so feature-level transmission is never activated where it would be wasteful; the fixed feature-level policies $C_{16}$ and $C_{256}$, by contrast, are strictly dominated—they cost 21–41× more channel use yet fall below Fixed $L$ in realised AP because of the LDPC cliff (Fig. 3). (iii) Averaged over all channel states the selector's channel use is 0.08102–0.20361 Msym/frame on validate, 0.03680–0.21196 on the scene-disjoint test split and 0.02437–0.18226 on Culver-City across the three budgets, i.e. 8.2–20.6%, 3.7–21.4% and 2.5–18.4% of Fixed $F$, because the expensive feature-active regime occupies only the high-SNR AWGN tail.

CA-TOSG is not designed to raise channel-averaged

TABLE III

Headline results: true end-to-end AP of CA-TOSG across three OPV2V splits, in the fading/low-SNR fallback regime (selector holds $L$) and the AWGN $\hat{\gamma}_t \geq 10$ dB feature-active regime (selector requests $F$), the boundary being the measured $\rho_F$ knee of Section VI-F, averaged over the six grid points at and above it. When the channel can carry features, CA-TOSG lifts AP on all three splits (+0.047 validate, +0.020 test, +0.018 Culver-City at IoU 0.5); when it cannot, the selector safely retains the $L$ operating point. Payload is per-frame channel use (Msym, at rate-1/2); channel-averaged CA-TOSG payload is 0.068–0.151 Msym/frame across splits, i.e. 7–15% of Fixed $F$ (0.990).

| Split | Regime | AP@0.5 | AP@0.7 | Payload |
|---|---|---|---|---|
| Validate (1,980) | Fallback ($L$) | 0.7818 | 0.7038 | 0.029 |
| | AWGN $\geq$ 10 dB | 0.8290 | 0.7296 | 0.483 |
| Test (2,170) | Fallback ($L$) | 0.8691 | 0.8027 | 0.024 |
| | AWGN $\geq$ 10 dB | 0.8892 | 0.8177 | 0.453 |
| Culver-City (550) | Fallback ($L$) | 0.7299 | 0.6463 | 0.024 |
| | AWGN $\geq$ 10 dB | 0.7475 | 0.6553 | 0.180 |

AP above Fixed $L$ on every frame: on many frames object-level communication is already sufficient, and a channel-averaged mean necessarily dilutes the gain. Its value is to activate feature-level communication only when the expected task gain justifies the payload and channel risk. The complementary channel-averaged payload–F1 frontier (Section VI-H), the difficulty stratification that localises the gain (Section VI-I), and the SNR-threshold comparison (Section VI-J) further characterise where and why the policy helps.

Because a one-line SNR-threshold rule is the natural challenger to any learned selector, we place it in the headline rather than an appendix. Table IV reports the channel-averaged frame F1 and per-frame channel use of Fixed $L$, a re-tuned SNR-threshold rule, the learned selector, and the oracle over 200 random SNR/channel draws on the frozen test split. The comparison is budget-indexed, and it does not run one way. At $B_{\max} = 0.10$ the selector's realised F1 is now marginally below that of a threshold tuned on validate for the same target budget, by −0.00099 (frame-level paired bootstrap 95% CI $[-0.00105, -0.00093]$) — the direction reversed under the corrected convention, and it is reported as it came out; this is a secondary comparison and we report the interval only, with no decision attached. At $B_{\max} = 0.20$ and 0.30 the nominal threshold rule attains the higher F1—0.89701 and 0.89900 against 0.89691 and 0.89783—while transmitting 1.53× and 1.47× the channel use, and doing so from outside the budget; against the budget-matched $\tau_{\text{feasible}}$ the F1 ordering at $B_{\max} = 0.20$ reverses in the selector's favour. The claim that survives is the pre-registered one, on the single confirmatory comparison (test at $B_{\max} = 0.20$): the selector's F1 is lower than the nominal threshold rule's by only ≈0.0001 (95% CI upper bound −0.00002, still entirely below zero) yet far within the pre-registered non-inferiority margin of 0.005, while reducing communication payload by 34.8% against a nominal threshold that is itself over budget (0.2168 > 0.20 Msym), or 26.6% against the budget-matched $\tau_{\text{feasible}}$. Frames within a scene are not independent, so we also resample that primary comparison at the level of the 16 test scenes (cluster bootstrap, 10,000 resamples): the F1 interval widens by 18.938× to $[-0.001508, +0.001522]$ and the payload interval by 45.177× to $[-0.119307, -0.029856]$ Msym. The scene-level F1 interval therefore no longer excludes zero, but it does not reach the pre-registered margin −0.005 either, and the payload advantage stays strictly negative; both intervals are reported together and the pre-registered inconclusive ruling is not triggered. On an LDPC + QAM cliff, in other words, channel state carries most of the accuracy decision — most, not all, and only on this transport — and what the ego-side cues buy is bandwidth, not peak accuracy. We report this rather than hide it: it is the central diagnostic finding of the paper, and it recurs as the codec-dependent currency of the task cues (Appendix A).

A qualitative example of such a frame—object-level F1 0.67 against feature-level 0.95 on the same frame, with the object-level branch producing false positives away from any ground-truth vehicle—is recovered by the repository's figure product (plot_qualitative_bev.py); the selector's job is to identify such frames and request $F$ only when the channel permits.

## B. Detection Performance versus SNR

Fig. 3 reports the frozen selectors' realised effective F1 against SNR under both channels, beside Fixed $L$, Fixed $F$, the perfect-channel feature ceiling and the feasibility-masked oracle. Every curve is read from the same frozen grid product (frozen_curves.csv); the codec-comparison baselines (importance-map JSCC and the LDPC + 16/256-QAM separate-coding arms) are a prior-protocol arm and are reported in Appendix A rather than mixed into this figure.

Under AWGN at $B_{\max} = 0.20$ the selector sits on the Fixed $L$ line (0.8554 on validate) for SNR ≤ 8 dB and steps to 0.8875 at 10 dB, where it stays flat to 20 dB. Two things about that step matter. It is a policy step, not a detection-quality step: it happens exactly where the 16-QAM frame-error cliff clears (Section III-C), and above it the curve is flat—there is no further AP to be had from a better channel. And it does not reach the ceiling: the perfect-channel feature reference is 0.8843 on validate and the masked oracle reaches 0.8927, so the

TABLE IV
Channel-averaged comparison on the frozen OPV2V test split (selector trained on validate, not retrained; $T = 2{,}170$ frames), read from the frozen replay: F1 and channel use are the means over the same 200 paired CSI samplings (SNR $\sim \mathcal{U}[0,20]$ dB, 50/50 AWGN/Rayleigh) for every row. The comparison is budget-indexed: each frozen selector is shown against the SNR-threshold rule tuned to the same budget, and it does not run one way. Share is channel use as a fraction of Fixed $F$ (0.990 Msym).

| Policy | Mean F1 | Channel use (Msym) | Share of Fixed $F$ |
|---|---|---|---|
| Fixed $L$ (object-level) | 0.8910 | 0.024 | 2.4% |
| Fixed $F$ (LDPC+16-QAM) | 0.8483 | 0.990 | 100.0% |
| Fixed $C_{256}$ (LDPC+256-QAM) | 0.8255 | 0.495 | 50.0% |
| Masked oracle (upper bound) | 0.9056 | 0.175 | 17.7% |
| $B_{\max} = 0.10$ | | | |
| SNR-threshold ($\tau^\star$=18) | 0.89247 | 0.0724 | 7.3% |
| CA-TOSG (RF, frozen) | 0.89148 | 0.0368 | **3.7**% |
| $B_{\max} = 0.20$ | | | |
| SNR-threshold ($\tau^\star$=12) | 0.89701 | 0.2168 | 21.9% |
| CA-TOSG (RF, frozen) | 0.89691 | 0.1414 | **14.3**% |
| $B_{\max} = 0.30$ | | | |
| SNR-threshold ($\tau^\star$=8) | 0.89900 | 0.3125 | 31.6% |
| CA-TOSG (RF, frozen) | 0.89783 | 0.2120 | **21.4**% |

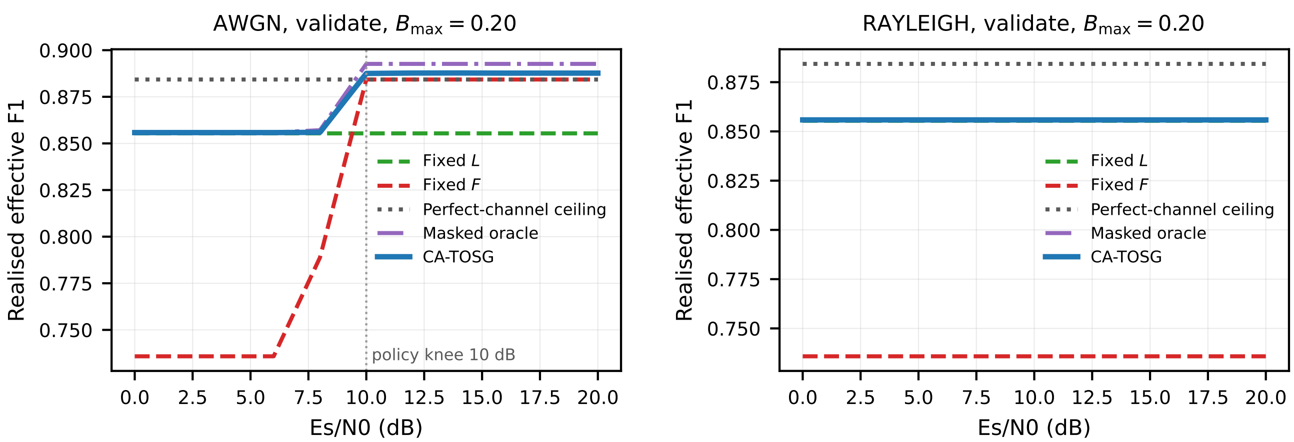


Fig. 3. Realised effective F1 versus SNR under AWGN (left) and Rayleigh (right) on OPV2V validate at $B_{\max} = 0.20$, every curve from the frozen grid. CA-TOSG holds $L$ below the cliff—its own realised F1 there is 0.8558, marginally above the Fixed-$L$ line at 0.8554 because of the $E/L$ mix—and steps to 0.8875 at the measured 10 dB knee, reaching 0.8877 by 20 dB; the perfect-channel feature ceiling is 0.8843 and the feasibility-masked oracle reaches 0.8927, so the step recovers part of the headroom rather than all of it. Under Rayleigh the selector stays on Fixed $L$ throughout, flat at 0.8558 across the whole range. The codec-comparison baselines are a prior-protocol arm and are in Appendix A.

frozen selector recovers part of the available headroom, not essentially all of it. Under Rayleigh the selector remains on Fixed $L$ across the entire SNR range; the deep-fade frame BLER never falls low enough for the feature branch to be worth requesting, even at 20 dB mean SNR. No fixed feature-level policy beats object-level communication on a channel-averaged basis—the low-SNR cliff outweighs the high-SNR gain—which is precisely the per-frame gating opportunity CA-TOSG exploits by holding $L$ below the threshold and activating $F$ above it.

Payload versus SNR. On the same frozen grid at $B_{\max} = 0.20$, CA-TOSG holds the $L$-payload (0.0237 Msym/frame on validate—the $E/L$ mix, measured at AWGN 0 dB and identical at every Rayleigh SNR) at every SNR under Rayleigh and for AWGN SNR $\leq 8$ dB, then steps once at 10 dB to 0.4795 Msym and stays there—about 48% of the Fixed-$F$ payload, not a near-$F$ payload: even with the channel fully open the frozen selector requests the feature message on only about a quarter of frames. The fixed-mode baselines are flat at their constant payload and give the worst-case bandwidth reference.

## C. Selector Decision Ratios

Fig. 4 shows the per-mode selection ratios $\rho_s$ against SNR, with the frozen selector overlaid on that budget's payload-penalised oracle and the stacked-area panel showing the share of each action of $\mathcal{S} = \{E, L, F\}$ at every SNR. Under AWGN the selector's adaptation knee is at 10 dB, where the frame-error cliff clears: $\rho_F$ steps from 0 to 0.472 on validate (0.433 on test, 0.160 on Culver-City) and is flat above it, with $\rho_L$ falling correspondingly. Under Rayleigh $\rho_L$ stays at essentially 1 across the whole range. The panel also makes the $E$-collapse of Section VI-H visible: under Rayleigh the oracle spends a substantial share on the ego-only action—$\rho_E = 0.157$ on test and 0.133 on Culver-City—while the frozen selector's $\rho_E$ is 0.002 and 0.000. $E$ is a deployed action that the imitation objective almost never selects, which is a limitation of the trained policy rather than of the action set. It is, however, not a limitation a simpler policy repairs: a hand rule with an explicit $E$ gate, fitted over three difficulty proxies in both orientations, selects $E$ in no split × budget cell ($\rho_E = 0.00000$ throughout, Section VI-K) — on these cues $E$ is structurally out of reach of a monotone threshold, and the frozen selector's small but non-zero $\rho_E$ is more than any of them attains. Under the corrected single-collaborator convention this limitation became cheaper without becoming milder. The behaviour is unchanged — the selector still almost never chooses $E$ — but the price of those missed calls fell from 0.58–0.61$\delta$ to 0.44–0.47$\delta$ on test, where $\delta = 0.005$ is the pre-registered non-inferiority margin. The oracle's own appetite for $E$ moved in opposite directions on the two splits: the number of grid cells it labels $E$ rose on validate (335 → 529) and fell on test (6,303 → 5,696), so the correction does not simply make ego-only more attractive everywhere.

## D. Feature Importance

Table V report the top Gini feature importances of the deployed selector. The two channel-side features dominate: the channel-type indicator $c_t$ contributes 34.2% of importance and the estimated SNR $\hat{\gamma}_t$ a further 27.5%, so two of the 23 inputs carry 61.7% between them against 38.3% spread over the 21 perception-side cues. The strongest of those, pcd_mean_range, reaches only 3.0%. Two cautions belong with that number. Importance is not sufficiency: a Gini share says how often a feature is split on, not that the remaining 38.3% is redundant, and the feature ablation shows the opposite — with the channel features alone the selector stops requesting features at the tighter budgets and never recovers the full selector's operating point (Section VI-E). And the split is transport-specific: it is measured under an LDPC + QAM cliff where the channel state is nearly decisive by construction; under graceful degradation the same cues carry the decision instead (Appendix A). That the channel-type indicator $c_t$ outranks the estimated SNR $\hat{\gamma}_t$ is consistent with the frame-level channel physics: under Rayleigh fading the feasible set collapses to $L$ regardless of SNR, so the channel-type flag is the single highest-information split for the decision.

## E. Ablation: Effect of Channel-State Features

To isolate the contribution of the two channel-state features, we train selector variants under identical RF hyperparameters over the feature subsets of Table VI: perception cues only (21 LiDAR cues); the perception cues plus the estimated SNR $\hat{\gamma}_t$; the full configuration adding the channel-type indicator $c_t$ (the deployed selector); and, for reference, channel state alone $(\hat{\gamma}_t, c_t)$ and a hand-tuned SNR threshold. Table VI summarises the results.

TABLE V
Top-7 features by Gini importance for the deployed CA-TOSG selector (trained on OPV2V validate, $T = 1{,}980$ frames). The two channel-side features account for 61.7% of total importance; the 21 perception-side cues carry the remaining 38.3%. Read directly from the frozen model, results/main/feature_importance_frozen.csv.

| Feature | Gini importance |
|---|---|
| channel_is_rayleigh | 0.342 |
| est_snr_db | 0.275 |
| pcd_mean_range | 0.030 |
| pcd_left_points | 0.028 |
| ego_num_objects | 0.025 |
| pcd_std_range | 0.023 |
| pcd_front_far_30m | 0.021 |
| $\hat{\gamma}_t + c_t$ subtotal | 0.617 |

TABLE VI
Selector feature ablation and the SNR-threshold challenger (train: validate, eval: frozen OPV2V test at $B_{\max} = 0.20$, $T = 2{,}170$ frames, 200 SNR/channel realisations). Each variant is retrained through the identical selection pipeline—same candidate set, same scene-level 9-fold LOSO, same frozen walk under the budget—with only the feature columns changed. Neither half of the input alone produces a graded policy: both the channel-only and the cues-only variants collapse to always transmitting $L$ at this budget (feature-request rate $\rho_F = 0$), so only the full 23-feature selector reaches an intermediate operating point. The cues-plus-$\hat{\gamma}$ cut of the earlier ablation was not re-run under the corrected protocol and is therefore not listed.

| Selector | #feat | Realised F1 | Deployed payload (Msym) |
|---|---|---|---|
| Channel only $(\hat{\gamma}, c)$ | 2 | 0.89095 | 0.02400 |
| Perception cues only | 21 | 0.89092 | 0.02392 |
| Full (all features) | 23 | 0.89691 | 0.14141 |
| SNR-threshold $(\tau^\star)$ | – | 0.89701 | 0.21679 |
| Fixed $L$ | – | 0.89095 | 0.02400 |

Two findings stand out. First, neither half of the input is sufficient on its own, and the failure mode is the same in both directions: run through the identical pipeline under $B_{\max} = 0.20$, the channel-only variant (2 features) and the cues-only variant (21 features) both collapse to always transmitting $L$—feature-request rate $\rho_F = 0.000$, payload pinned at $B_L$, realised F1 at the fixed-$L$ floor 0.8909—while the full selector reaches 0.8969 at 0.141 Msym. Cues without a channel signal cannot tell whether a feature request would survive the link; channel state without cues cannot tell whether it would be worth sending. Only the combination yields a policy that sits between the two extremes. The collapse has a single mechanism, and it explains the same behaviour wherever it appears in this paper. The committed frame BLER is effectively binary in the channel state—1.0 for Rayleigh at every tabulated SNR, and for AWGN below 8 dB, against 0.0 at and above 10 dB—so a policy whose feature decision depends on the channel alone can realise only four mean payloads, 0.024, ≈0.27, ≈0.33 and 0.99 Msym, and the two tighter budgets fall between those rungs. That is why the channel-only variant pins itself to $B_L$ at $B_{\max} = 0.10$ and 0.20 and only activates $F$ at 0.30; why the pre-registered two-scalar hand rule degenerates to Fixed $L$ at the same

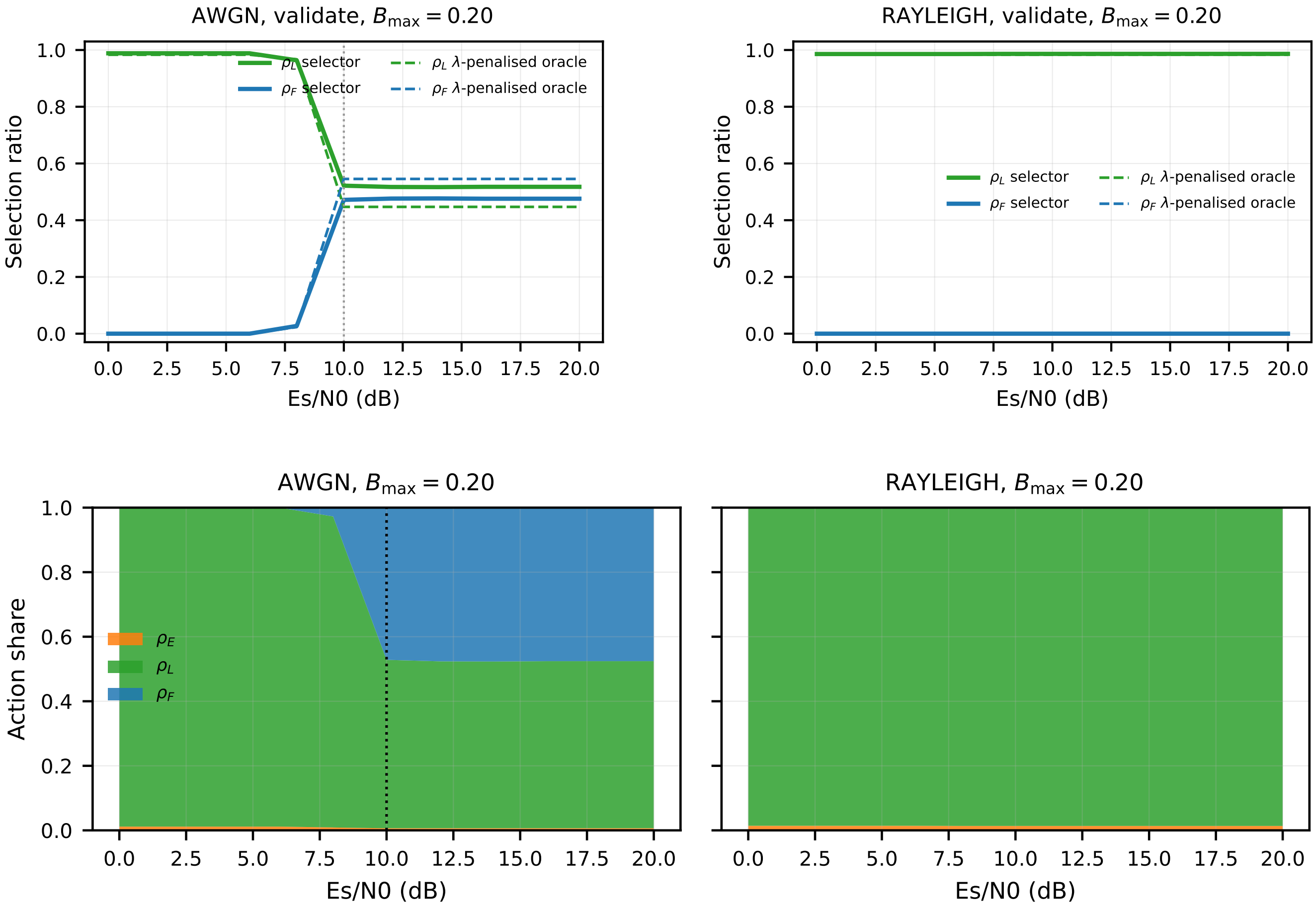


Fig. 4. Per-mode selection ratios versus SNR on OPV2V validate at $B_{\max} = 0.20$, from the frozen grid. Top: the frozen selector (solid) against that budget's $\lambda$-penalised oracle (dashed) — the oracle sees the frame BLER, not the per-frame block outcome, so it is not clairvoyant. Bottom: stacked-area view of the selector's action distribution over the deployed action set $\mathcal{S} = \{E, L, F\}$ under AWGN and Rayleigh (the dominated $C_{256}$ is a physical-layer comparator, not a deployed action, Section III-B); the dotted line marks the measured 10 dB policy knee. Two readings: under AWGN $\rho_F$ steps from 0 to 0.472 at 10 dB and is flat above it, and under Rayleigh the selector's $\rho_E$ is 0.002 (test) / 0.000 (Culver-City) while the oracle spends $\rho_E = 0.157$ / 0.133 there — the $E$-collapse limitation of Section VI-H. The three-budget version is the per-budget action distributions (repository figure product).

two budgets (Section VI-K); and why the nominal SNR threshold lands over budget at $B_{\max} = 0.20$ rather than on it. Reaching an intermediate budget requires deciding which frames deserve $F$ among those whose link can carry one, which is a per-frame content decision and is exactly what the perception cues supply. This does not contradict the sufficiency of the SNR threshold in Section VI-J: SNR is decisive as the threshold signal that gates the feature request under AWGN, but as an additional RF feature atop the perception cues it adds a noisy continuous dimension without an F1 gain—the two roles are distinct. Second, what the perception cues buy is deployed channel use rather than accuracy, and the effect is budget-dependent. At $B_{\max} = 0.10$ and 0.20 the channel-only variant has no graded policy to offer at all ($\rho_F = 0$, payload pinned at $B_L$). At $B_{\max} = 0.30$ it does activate the feature action ($\rho_F = 0.274$), but under the corrected convention it is now beaten on both axes at once: on test it reaches 0.89529 against the full selector's 0.89783 ($-0.0025$ F1) while spending 0.287 Msym against 0.212—1.36× the channel use for less accuracy. Under the retired accounting this variant bought +0.0021 F1 for that extra spend; the correction removes even that. Four independently constructed variants collapse in the same way on this substrate — channel-state-only, cues-only, a contextual-bandit comparator retrained on the corrected grid (which selects one $\lambda$ for all three budgets and settles at a near-$L$ operating point, 0.0294 Msym), and the SECOND-backbone selector of Appendix B — and only the full-feature imitation selector produces three distinct, budget-indexed operating points. The comparison is about which construction yields a graded policy, not about one method outperforming another. Under the corrected convention the cues do add accuracy on this axis (+0.0090 over channel state alone); what they mainly buy is still that comparable accuracy can be bought more cheaply.

TABLE VII
True end-to-end AP under the frozen selector on OPV2V validate ($T$ = 1,980 frames) at $B_{\max} = 0.20$, on the pre-registered SNR grid with 200 paired realisations per point, scored by the same engine as Tables IX and X. The feature-active regime begins at the measured $\rho_F$ knee (10 dB, Section VI-F), not at a stipulated threshold. AWGN rows only: under Rayleigh the frozen selector requests $F$ at no SNR ($\rho_F = 0$ throughout, Fig. 4), so every Rayleigh row is identically the 0 dB value.

| Channel | SNR (dB) | $\rho_L$ | AP@0.5 | AP@0.7 |
|---|---|---|---|---|
| AWGN | 0.0 | 0.988 | 0.7816 | 0.7040 |
| AWGN | 8.0 | 0.964 | 0.7827 | 0.7028 |
| AWGN | 10.0 | 0.522 | 0.8287 | 0.7292 |
| AWGN | 12.0 | 0.517 | 0.8291 | 0.7297 |
| AWGN | 16.0 | 0.518 | 0.8290 | 0.7297 |
| AWGN | 20.0 | 0.518 | 0.8290 | 0.7296 |

## F. True End-to-end AP Verification

The realised frame F1 reported so far is computed under the analytical block-loss model of Eq. (6), where a lost feature block falls back to the ego vehicle's own pre-fusion detection. To confirm that this analytical F1 proxy translates to actual end-to-end detection AP under a deployed pipeline, we run a true end-to-end inference in which: (i) the deployed selector picks $s_t$ per frame from $(\hat{\gamma}_t, c_t)$; (ii) cached per-frame predictions from the OPV2V late-fusion checkpoint (for $L$) and the attentive-compression checkpoint (for $C_q$) are concatenated according to the selector's choice; (iii) each $C_q$ frame is dropped with probability $\mathrm{BLER}_q(\gamma_t, \mathrm{ch}_t)$ and falls back to the ego-only (pre-fusion) prediction; (iv) the resulting box set is scored against ground-truth boxes with the OpenCOOD VOC-style AP computation at IoU thresholds $\{0.3, 0.5, 0.7\}$, averaged over the 200 paired stochastic channel realisations of the frozen replay. Table VII reports the result.

Two observations support the central claim. First, the boundary of the feature-active regime is measured, not stipulated: reading the frozen selector's feature-request rate off the pre-registered SNR grid, $\rho_F$ is zero at every AWGN point up to 8 dB and steps to its plateau at 10 dB on all three splits—0.472 on validate, 0.433 on test and 0.160 on Culver-City—which is exactly where the 16-QAM frame-error cliff clears (Section III-C).[3] At that boundary the true end-to-end AP on validate is 0.8287 at IoU 0.5 and 0.7292 at IoU 0.7, and it is flat above it. The selector reaches this while paying 0.300 Msym/frame averaged over the feature-active regime instead of the 0.990 of always-on Fixed $F$—a 70% saving—though on validate it recovers only part of the perfect-channel reference of the underlying attentive-compression model (0.8369 on validate, Section VI-A) rather than essentially all of it, as an earlier version of this section claimed. Second, the Rayleigh AP curve is flat across the full SNR range and matches Fixed $L$, confirming that the selector's conservative fading-regime behaviour, identified in Section VI-A, is not a frame-F1 artefact but a genuine end-to-end property.

[3]Validate shows a 0.4% toe at 8 dB, the single grid point at which the frame BLER is neither ≈0 nor ≈1 (0.402); test and Culver-City are exactly zero there. We therefore take 10 dB as the regime boundary on all three splits and report the toe rather than smoothing it away. Earlier versions of this section stipulated ≥ 14 dB; that threshold was not derived from any measurement and is withdrawn.

## G. Generalisation to OPV2V Test and Culver-City Splits

All results above are computed on the OPV2V validate split. To test whether the selector overfits to that split, we re-run the entire pipeline—per-frame late-fusion and attentive-compression inference, the ego-side cues, the channel-aware oracle labelling, and the true end-to-end AP scoring—on two held-out splits never seen during selector training: (a) the OPV2V test split ($T$ = 2,170 frames, scene-disjoint from validate), and (b) the OPV2V Culver-City split ($T$ = 550 frames), a domain shift in which the simulated scenes follow the real road network of Culver City rather than the default CARLA towns used in the other splits. Crucially, the deployed Random Forest selector is not retrained on either split: the exact validate-trained model is applied unchanged, so this measures genuine cross-split and cross-domain generalisation rather than re-fitting.

Table VIII reports the headline comparison on both splits, every row read from the frozen replay. The meaningful evidence of transfer is that the policy ordering is identical to validate on both splits: the fixed feature-level policies remain dominated by Fixed $L$, the masked oracle sits marginally above Fixed $L$, and each frozen selector sits between them. Quantitatively the transfer is in the channel use, not the F1: across the three budgets CA-TOSG spends 3.7–21.4% of the Fixed-$F$ channel use on test and 2.5–18.4% on Culver-City, while its realised F1 stays within 1.6 percentage points of the masked oracle's on test (98.4–99.1% of it) and within 2.0 on Culver-City (98.0–99.4%).

Tables IX and X report the true end-to-end AP on the test and Culver-City splits, evaluated with the frozen selector at $B_{\max} = 0.20$ on the pre-registered 11-point SNR grid, 200 paired realisations per point. Three things follow, and only one of them is the behaviour an earlier version of this section reported. First, the knee is real but it is a policy knee, not an AP knee, and it sits at 10 dB—exactly where the 16-QAM frame-error cliff clears (Section III-C): $\rho_F$ jumps from 0 to 0.433 on test and from 0 to 0.160 on Culver-City, and it is flat thereafter. Second, that policy change buys a small but non-zero AP gain, and on test it costs a little: AP@0.5 moves 0.8691 → 0.8896 and AP@0.7 0.8027 → 0.8187 on test, while on Culver-City it moves 0.7299 → 0.7474 and 0.6463 → 0.6556—gains of 2.0 and 1.8 AP@0.5 points, larger than the sub-point moves the retired accounting showed but still small against the 2.4 and 9.7 point headroom. This is the same reading as the headline: at these budgets the frozen selector requests $F$ sparingly, and what it transfers across the domain shift is the communication saving rather than an AP gain. Third, the Rayleigh curves are flat at the Fixed-$L$ operating point across the entire 0–20 dB range on

TABLE VIII
Generalisation of the validate-trained frozen selectors (no retraining) to the OPV2V test ($T = 2{,}170$ frames, scene-disjoint) and Culver-City ($T = 550$ frames, real-road-layout domain shift) splits. Every row is the mean over the same 200 paired CSI samplings of the frozen replay (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh); each frozen selector is paired with the SNR-threshold rule tuned to the same budget. Share is channel use as a fraction of Fixed $F$ (0.990 Msym).

| Policy | Mean F1 | Channel use (Msym) | Share of Fixed $F$ |
|---|---|---|---|
| OPV2V test ($T = 2{,}170$) | | | |
| Fixed $L$ | 0.8910 | 0.024 | 2.4% |
| Fixed $F$ (LDPC + 16-QAM) | 0.8483 | 0.990 | 100.0% |
| Fixed $C_{256}$ (LDPC + 256-QAM) | 0.8255 | 0.495 | 50.0% |
| CA-TOSG $B_{\max}$=0.10 / $\tau^\star$ | 0.89148 / 0.89247 | 0.0368 / 0.0724 | 3.7% / 7.3% |
| CA-TOSG $B_{\max}$=0.20 / $\tau^\star$ | 0.89691 / 0.89701 | 0.1414 / 0.2168 | 14.3% / 21.9% |
| CA-TOSG $B_{\max}$=0.30 / $\tau^\star$ | 0.89783 / 0.89900 | 0.2120 / 0.3125 | 21.4% / 31.6% |
| Channel-aware oracle (masked) | 0.9056 | 0.175 | 17.7% |
| OPV2V Culver-City ($T = 550$) | | | |
| Fixed $L$ | 0.8466 | 0.024 | 2.4% |
| Fixed $F$ (LDPC + 16-QAM) | 0.8099 | 0.990 | 100.0% |
| Fixed $C_{256}$ (LDPC + 256-QAM) | 0.7793 | 0.495 | 50.0% |
| CA-TOSG $B_{\max}$=0.10 / $\tau^\star$ | 0.87230 / 0.87491 | 0.0283 / 0.0718 | 2.9% / 7.2% |
| CA-TOSG $B_{\max}$=0.20 / $\tau^\star$ | 0.87355 / 0.88340 | 0.0423 / 0.2174 | 4.3% / 22.0% |
| CA-TOSG $B_{\max}$=0.30 / $\tau^\star$ | 0.88286 / 0.88740 | 0.1601 / 0.3146 | 16.2% / 31.8% |
| Channel-aware oracle (masked) | 0.8643 | 0.266 | 26.8% |

TABLE IX
True end-to-end AP on the OPV2V test split under the frozen validate-trained selector at $B_{\max} = 0.20$, 200 paired realisations per operating point, on the pre-registered SNR grid. The knee at 10 dB is a policy knee ($\rho_F$: $0 \to 0.433$) that coincides with the 16-QAM frame-error cliff; AP rises by 2.0 AP@0.5 points across it and is flat thereafter. AWGN rows only: under Rayleigh the frozen selector requests $F$ at no SNR ($\rho_F = 0$ throughout, Fig. 4), so every Rayleigh row is identically the Fixed-$L$ value 0.8691 / 0.8027.

| Channel | SNR (dB) | $\rho_L$ | AP@0.5 | AP@0.7 |
|---|---|---|---|---|
| AWGN | 0.0 | 0.998 | 0.8691 | 0.8027 |
| AWGN | 8.0 | 1.000 | 0.8691 | 0.8027 |
| AWGN | 10.0 | 0.567 | 0.8896 | 0.8187 |
| AWGN | 12.0 | 0.555 | 0.8891 | 0.8176 |
| AWGN | 16.0 | 0.554 | 0.8892 | 0.8175 |
| AWGN | 20.0 | 0.553 | 0.8891 | 0.8174 |

TABLE X
True end-to-end AP on the OPV2V Culver-City split (real-road-layout domain shift, $T = 550$ frames) under the frozen validate-trained selector at $B_{\max} = 0.20$, 200 paired realisations per operating point. Same 10 dB policy knee ($\rho_F$: $0 \to 0.160$), a +1.8 AP@0.5 point change across it. AWGN rows only: under Rayleigh the frozen selector requests $F$ at no SNR ($\rho_F = 0$ throughout, Fig. 4), so every Rayleigh row is identically the Fixed-$L$ value 0.7299 / 0.6463 and carries no information beyond it. Absolute AP is lower than on test because the domain is harder.

| Channel | SNR (dB) | $\rho_L$ | AP@0.5 | AP@0.7 |
|---|---|---|---|---|
| AWGN | 0.0 | 1.000 | 0.7299 | 0.6463 |
| AWGN | 8.0 | 1.000 | 0.7299 | 0.6463 |
| AWGN | 10.0 | 0.840 | 0.7474 | 0.6556 |
| AWGN | 12.0 | 0.836 | 0.7476 | 0.6554 |
| AWGN | 16.0 | 0.836 | 0.7476 | 0.6554 |
| AWGN | 20.0 | 0.840 | 0.7473 | 0.6551 |

both splits, because the deep-fade frame BLER never falls low enough to make feature-level transmission worthwhile; this is the one part of the earlier account that reproduces unchanged. The non-monotonic dip previously reported at the 20 dB training-grid edge does not appear under the frozen selector—AP is flat above 10 dB on both splits—so the grid-edge artefact explanation is withdrawn.[4]

### H. Communication–Perception Pareto Frontier

The headline table compares policies at a single operating point. To show that CA-TOSG sits on the attainable payload–F1 frontier rather than merely better on average, we sweep the Lagrangian selector $s_t^\star = \arg\max_s (F_t^s - \lambda B_s)$ over $\lambda$ and plot every policy on the payload–F1 plane (Fig. 5). As $\lambda$ grows from 0 (max F1) to large values (min payload), the oracle traces the achievable upper frontier; the fixed feature-level policies $C_{16}$ and $C_{256}$ sit far below it (high payload, low realised F1 due to the LDPC cliff), i.e. they are strictly dominated. CA-TOSG and a $\lambda$-cost RF retrained on $\arg\max_s (F_t^s - \lambda B_s)$ labels lie on the frontier between Fixed $L$ and the oracle. The P2 product is three frozen selectors, one per budget, so the operating point is budget-indexed: on test they realise F1 0.89148, 0.89691 and 0.89783 at 0.037, 0.141 and 0.21196 Msym/frame for $B_{\max} = 0.10$, 0.20 and 0.30. At $B_{\max} = 0.20$ that payload is 14.3% of the feature-level payload $B_F$.

### I. Where the Gain Concentrates: Difficulty Stratification

A channel-averaged mean masks where CA-TOSG matters. We stratify frames into easy/medium/hard terciles by the ego's own object-level detection F1 (low = hard) and evaluate the frozen selectors at a single reliable-channel operating point—deterministic AWGN at 16 dB,

[4]The per-SNR curves of an earlier version of this section were produced by the retired global-sort scorer and its selector, not by the frozen selectors, and are withdrawn rather than restated. The frozen replacement (end_to_end_ap_snr.py) differs from the deployed AP engine in exactly one respect—the SNR/channel draw is pinned instead of sampled—and is gated on reproducing every committed row of the deployed table before any pinned number is produced. Coverage: test and Culver-City, AWGN and Rayleigh, all three frozen budgets; validate is not re-run here.

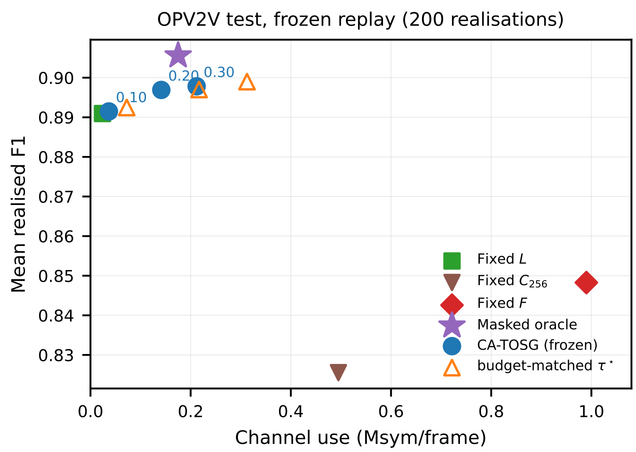


Fig. 5. Communication–perception plane on OPV2V test, from the frozen 200-realisation replay. The axis spans the full range so Fixed $F$ (0.99 Msym) is visible rather than clipped. The three frozen selectors are shown budget-indexed (circles, labelled by $B_{\max}$) beside the SNR-threshold rule tuned to the same budget (open triangles); the feasibility-masked oracle is a separate marker (star). Fixed $F$ and Fixed $C_{256}$ are dominated by Fixed $L$ at far higher channel use.

one point of the pre-registered SNR grid rather than a channel draw (the difficulty stratification (repository figure product)). At $B_{\max} = 0.20$ the CA-TOSG-over-Fixed-$L$ gain rises monotonically with difficulty: on test it is $-0.0047$ (95% CI $[-0.0074, -0.0024]$) on easy frames, $+0.0056$ ($[+0.0027, +0.0085]$) on medium frames and $+\mathbf{0.0660}$ ($[+0.0591, +0.0730]$) on hard frames. Validate has the same shape ($+0.0058/+0.0441/+0.0470$); Culver-City is weaker throughout ($+0.0011/+0.0021/+0.0310$). The value of CA-TOSG is therefore two-dimensional: it materialises where a hard frame meets a reliable channel, and a channel-averaged headline necessarily dilutes it. On easy frames the deployed selector over-requests $F$ on test, and the frozen product puts the cost at $-0.00471$ F1 (95% CI $[-0.00742, -0.00236]$) at $B_{\max} = 0.20$, growing monotonically with the budget —$-0.00020$, $-0.00471$, $-0.01129$ across 0.10/0.20/0.30—as the looser cap lets more easy frames be sent as $F$. The effect is split-dependent rather than universal: on validate and Culver-City the same stratum gains at every budget ($+0.00035$ to $+0.01016$ and 0.00000 to $+0.00558$), so it is a property of the test split's easy tercile, where fixed $L$ already reaches 0.97962 and there is almost nothing left to buy. A payload-penalised ($\lambda > 0$) operating point on the Lagrangian frontier of Section VI-H would mitigate it.[5]

### J. Is a Learned Selector Necessary? Comparison with an SNR-Threshold Rule

Since the channel features are the individually strongest cues (Section VI-D), we test whether a hand-tuned rule suffices. Table VI compares RF feature subsets and a rule "if AWGN and $\hat{\gamma}_t > \tau$ then $F$ else $L$" (train on validate, evaluate on test). We report this honestly. Under cliff transport a retuned SNR threshold captures most of the attainable F1: on the payload–F1 frontier the learned RF and a tuned threshold sit almost on top of each other. As established in Section VI-A, the selector's edge is not on F1: on test the nominal threshold rule attains the marginally higher realised F1 at every budget—0.89247, 0.89701 and 0.89900 against the selector's 0.89148, 0.89691 and 0.89783—while spending 1.97×, 1.53× and 1.47× the channel use, and at $B_{\max} = 0.20$ and 0.30 doing so from outside the budget. Against the budget-matched $\tau_{\text{feasible}}$ the ordering reverses at one budget only: the selector is ahead by $+0.00067$ at $B_{\max} = 0.20$ and behind by $-0.00174$ and $-0.00118$ at 0.10 and 0.30. What is pre-registered and survives is non-inferiority within a 0.005 margin at a 34.8% payload reduction — a figure that must be read with the comparator's own budget in view: the re-tuned threshold spends 0.2168 Msym against the $B_{\max} = 0.20$ cap, i.e. it is over budget, so part of that reduction is the comparator's overspend rather than the selector's thrift. Against $\tau_{\text{feasible}}$, the strictly budget-matched threshold of Section VI-J ($\tau = 13$, 0.1927 Msym), the reduction is 26.6% and the selector is ahead on F1 by $+0.00067$ rather than behind. Neither half of the input is sufficient on its own, and the shape of the failure is the informative part: given only the channel state the selector stops requesting features altogether at the two tighter budgets (feature-request rate 0, payload pinned at $B_L$), and at $B_{\max} = 0.30$, where it does activate the feature action, it is beaten on both axes at once—0.89529 against the full selector's 0.89783 while spending 1.36× the channel use (Section VI-E). We attribute the small accuracy margin to the sharp SNR cliff: under this transport the channel state carries most of the decision, and a one-line threshold on it therefore tracks the learned selector closely. That is a statement about this transport, not about sufficiency in general — the same cues become decisive as soon as the codec degrades gracefully (Appendix A), and channel-side importance is not the same thing as channel-side sufficiency. The sentence this section settles on, and the one the rest of the paper is written to, is therefore: CA-TOSG does not win by raising average F1. A one-line SNR threshold and a three-scalar hand rule reach comparable F1, but they do so by sending more feature messages. What the task cues buy is knowing which of the frames the channel can carry are actually worth sending, so that perception stays non-inferior at lower communication. We therefore frame CA-TOSG as a lightweight, interpretable channel-aware selection policy whose aggregate operating point tracks a budget-matched threshold on this cliff codec closely, edging it on F1 while spending 26.6% less; the perception cues become decisive on graceful-degradation channels, where SNR is no longer a sufficient statistic and, in-distribution, the cue-based selector beats the best SNR threshold using the task cues alone (Appendix A). The channel-averaged payloads are

[5]The channel-averaged companion view reported in an earlier version of this subsection is withdrawn rather than recomputed: it was a product of the retired 200-realisation policy engine and its selector, not of the frozen selectors evaluated here, and the two are different quantities that may not be placed side by side.

in Table IV. Against the re-tuned SNR-threshold rule on test at $B_{\max} = 0.20$, the selector's F1 is lower than the nominal threshold rule by ≈0.0001 (95% CI upper bound −0.00002, so still entirely below zero) yet within the pre-registered non-inferiority margin of 0.005, while reducing communication payload by 34.8% against a threshold that is itself over budget (0.2168 > 0.20 Msym), or 26.6% against the budget-matched $\tau_{\text{feasible}}$. At $B_{\max} = 0.20$ and 0.30 the nominal threshold rule attains a marginally higher realised F1 (0.89701 and 0.89900 versus 0.89691 and 0.89783) at 1.53× and 1.47× the payload; against $\tau_{\text{feasible}}$ the ordering reverses at $B_{\max} = 0.20$ (0.89691 versus 0.89624).

## K. Hand-Rule Baselines and Their Communication Cost

A hand rule with a second gate is the next challenger after a one-parameter threshold, and was pre-registered before it was run: $a_t = E$ if $d_t \leq \tau_E$, else $F$ if $r_t = 1 - \mathrm{BLER}_F(\hat{\gamma}_t, c_t) \geq \tau_F$, else $L$, with $d_t$ one of three difficulty proxies from the same 21 cues in both orientations, both thresholds fitted on validate through the identical LOSO walk and budget constraint as the deployed selector.

At the two tighter budgets it cannot bid at all: the walk selects never-$E$ and never-$F$ at $B_{\max} = 0.10$ and 0.20, degenerating to Fixed $L$, and at 0.30 the fitted reliability gate rediscovers the SNR threshold ($\rho_F = 0.29853$ against 0.29883 on validate). This is the four-rung ladder of Section VI-E: gating on the channel alone reaches only four mean payloads, and 0.10 and 0.20 fall between rungs.

Because a comparator that cannot enter the comparison is not a comparator, the arm was amended—in the record, before running—so the feature gate is a conjunction: $F$ when $d_t \geq \tau_D$ and $r_t \geq \tau_F$. That is three scalars, and we report it as three. It is feasible at every budget and at the primary cell it matches the selector on F1: 0.89697 against 0.89691, paired difference +0.00005 (95% CI $[-0.00002, +0.00012]$). No further comparator was run after seeing it.

The difference lies on the other axis: the hand rule spends 0.19900 Msym against 0.14141, a paired difference of +0.05760 (95% CI $[+0.05663, +0.05853]$)—40.7% more channel for that +0.00005 F1, or the selector reaching the same F1 with 28.9% less. The mechanism is selectivity per transmitted frame: over the cells each policy sends $F$ on, the mean feature-over-object gain is 0.05022 (selector), 0.03660 (hand rule) and 0.03038 (SNR threshold)—the last indistinguishable from the 0.03042 unconditional mean, so the threshold's feature frames are no better chosen than average. The selector buys more F1 per transmitted frame and needs fewer of them; that is the payload gap, and it is what the ego-side cues contribute here.

Neither arm ever selects $E$: $\rho_E = 0.00000$ in every split × budget cell of both the two-scalar and the three-scalar rule, over all six cue orientations (Section VI-C).

## L. Collaboration Is Not Always Beneficial

Collaboration is not unconditionally beneficial, and that is what makes an explicit per-frame selector necessary rather than a convenience. Two failure modes call for two responses. When the channel cannot carry a feature message, requesting one spends the collaborator's transmission budget for nothing and collapses the output to the ego-only floor (the ego vehicle's own pre-fusion detection); the design answer is already in place – the oracle removes any action whose frame-level failure probability is ≥ 0.999 from its feasible set (the same 0.999 constant as the §IV mask), and on failure the pipeline reverts to the ego-only output rather than a phantom feature. When the channel can carry the message, requesting it can still cost accuracy: on the easy stratum the selector's realised output falls below even the always-object-level (Fixed-$L$) baseline.[6] This mode has no masking answer; its remedy is left to future work. Two CSV-verified quantifiers bound where the ego-side harm sits: the ego-only output strictly exceeds the object-level fused output on 1.5 / 5.8 / 0.2% of validation / test / Culver-City frames, and – from the same per-frame (comp − ego) identity, re-derived on the corrected single-collaborator caches – the compressed-feature message, when delivered, yields lower frame F1 than the ego-only fallback on 1.3 / 6.5 / 0.9% of frames. Test carries the harm most, consistent with fusion having the least to add in thin scenes (mean 15.2 ground-truth objects on test vs 27.8 on validate and 41.0 on Culver-City). No new signalling is needed to act on this: $E$ (ego-only, issue no request) is already a deployed action of $\mathcal{S}$ with its own codepoint, so the remedy is a selection-policy question—the frozen selectors almost never choose it (Section VI-H)—not a message-format one.

## M. Comparison with Where2comm

To place CA-TOSG beside the most prominent spatial-selection cooperative perception method, we report a reference-only comparison against Where2comm [5] on the same OPV2V validate split (1,980 frames). Three caveats bound what it can support, and they are stated rather than buried: the numbers come from our own 50-epoch reproduction, not the authors' released model; they are measured under a perfect channel, so no part of this paper's transport model applies to them; and they were scored by the retired global-sort scorer, which is a different track from the frozen AP chain that produces every headline number here. Our reproduction reaches

[6] Test Easy stratum (top tercile of the ego's own object-level F1, the §VI-I stratification), evaluated under a deterministic reliable-channel condition (AWGN 16 dB; frame-level BLER ≈0, well above the 8.0 dB onset), isolating the difficulty axis from channel variability: the frozen selector's realised F1 at $B_{\max} = 0.20$ is 0.9749 vs the Fixed-$L$ baseline 0.9796 – a gain of −0.0047 (frame-level paired 95% CI $[-0.0074, -0.0024]$; $n = 713$ frames). The selector requests $F$ on 241 of these 713 frames; on the remaining 472, where it requests $L$, its output is frame-identical to Fixed-$L$, so the paired difference arises entirely on the $F$-request frames – the loss is a structural consequence of requesting features on already-easy frames.

AP@0.5 = 0.871 under a perfect channel, and we deliberately draw no ranking from it. It is not comparable to any number in this paper, on three independent axes at once: a different scorer (the retired global-sort track, not the frozen AP chain), a perfect channel rather than the modelled transport, and—decisively—a different collaborator convention, since the reproduction fuses every collaborator while the corrected main experiment pays for one. The direction of such a comparison is not even stable: under the corrected accounting the validate Fixed-$L$ reference is 0.7819, below 0.871, which reverses the ordering an earlier version of this section reported. Numbers measured under three different conventions cannot be ordered, and we do not order them.

The comparison is also orthogonal to CA-TOSG: Where2comm is a spatial-mask method that decides which feature regions to send within a fixed feature-level granularity, whereas CA-TOSG decides whether to send feature-level information at all and at what granularity, given the channel. The two compose: Where2comm can be deployed inside the $C_q$ branches of CA-TOSG as an alternative feature codec, while the granularity selector decides when feature-level communication is worth its payload and channel risk. Under a bandwidth-constrained, time-varying link, even a feature-level message is realised only when the channel can deliver it—precisely the regime CA-TOSG arbitrates.

## N. Collaborator Scale

The evaluation so far treats the collaborator set as given. To ask how the policy behaves as more vehicles are reachable, we sweep the number of requested collaborators $N \in \{1, 2, 3\}$, keeping the frozen selector, $\lambda^\star$ and $\tau^\star$ untouched.

The first thing the sweep exposes is that $N$ is not the operative quantity. What matters is the effective collaborator count $k_{\text{eff}}$—how many collaborators a frame actually has—because requesting three cannot conjure a third. Of the 1,980 validate frames, 733 are already saturated at $N = 1$, 1,081 at $N = 2$ and 1,234 at $N = 3$; on Culver-City the supply runs out entirely, so $N = 2$ and $N = 3$ are the same arm and we report them as one. All comparisons below are therefore conditioned on the same frames rather than taken across differently-sized frame populations.

Under that accounting, F1 rises with $k_{\text{eff}}$ with sharply diminishing returns while channel use grows sub-linearly. At $B_{\max} = 0.20$ on validate, going from $N = 1$ to $N = 2$ buys +0.0300 F1 for +0.1183 Msym, and the third collaborator adds only +0.0061 more F1 for a further +0.0991 Msym—about a fifth of the gain for three-quarters of the extra channel use. On test the pattern is the same but flatter (+0.0096 then +0.0008), because that split is thinner and saturates earlier. The sub-linearity in payload is a supply effect rather than a policy effect: the per-frame action distribution is unchanged across $N$ (the frozen selector sees the same cues), so the extra channel use comes only from the frames that actually gained a collaborator.

Because “request $N$” is ambiguous when fewer than $N$ collaborators exist, we bracket the two delivery semantics—charging only for what is delivered, versus charging for what was requested—on validate at $N = 2$, on the 964 frames where the two differ. The bracket is tight: charging for what is delivered rather than for what was requested raises realised F1 by +0.00016, +0.00036 and +0.00099 across the three budgets, at identical payload—the request is what is charged under both—so the conclusions here do not depend on which convention is adopted. Finally, the budget does not transfer automatically. The selector is frozen against an average per-frame budget measured with a single collaborator; total communication grows with the number of collaborators actually served, so the original budget guarantee does not carry over unchanged. This is measurable rather than hypothetical: on validate at $B_{\max} = 0.20$, $N = 3$ spends 0.36842 Msym per frame, well above the 0.20 it was frozen to satisfy — and under the corrected single-collaborator accounting even $N = 2$ (at 0.26937) already breaches it. A deployment that serves more collaborators must therefore re-freeze, or budget on $k_{\text{eff}}$ rather than per collaborator.

## O. Deployment Robustness and Cost

Three practical imperfections—a noisy SNR estimate, channel-state aging, and a stale request—degrade the selector only gracefully, because an uncertain channel state falls back to $L$ (Table XI): SNR-estimation noise up to $\sigma \approx 1$ dB costs $\leq 0.0002$ F1; a Jakes model at 60 km/h decorrelates the estimate within ∼10 ms, costing −0.0025 before plateauing; a one-frame-stale decision costs at most −0.0109 under our i.i.d.-per-frame assumption (changing 18.2% of decisions). The near-immunity to estimation noise follows from the channel-type-anchored policy of Section VI-D: perturbing continuous $\hat{\gamma}_t$ rarely flips a decision anchored on the discrete channel-type flag, and under Rayleigh the action remains $L$ regardless. Finally, the deployed Random Forest runs in 52.1 ± 5.6 ms per frame (P95 = 58.3 ms) on a single CPU core, i.e. the selector's own inference occupies about half of the 100 ms frame interval of a 10 Hz LiDAR cycle; we measure the selector only and make no claim about the end-to-end perception–transmission chain. The figure is the slowest of the three frozen selectors over 1,000 batch-1 trials each; the deployment claim must hold for every budget, so the worst case is the one quoted.

A fourth imperfection concerns the cheap message rather than the expensive one. The utilities above model the object-level link as reliable ($\text{BLER}_L = 0$), which is an assumption about the low-rate path, not a measurement. Re-weighting the object-level utility analytically as $\text{eff}'_L = \text{late}\,(1 - \text{BLER}_L) + \text{ego}\,\text{BLER}_L$ over $\text{BLER}_L \in \{0.01, 0.05, 0.10\}$, with every policy frozen and blind to it, leaves the payload — and therefore the payload advantage — untouched by construction, and costs F1 roughly linearly: on test at

TABLE XI
Deployment robustness (OPV2V test; realised frame F1). Each row perturbs one imperfection while holding the others ideal; losses are bounded because an uncertain channel state defaults to the robust $L$ message. Re-derived under the single-collaborator convention with the frozen selectors at $B_{\max} = 0.20$ (results/sensitivity/robustness_frozen.csv); the CSI-noise and Jakes models are the ones used previously, so only the selector, grid and utilities changed.

| Imperfection | Operating point | ΔF1 |
|---|---|---|
| SNR-estimation noise $\sigma$ | $\leq 1$ dB | −0.0002 |
| | 2 dB | −0.0006 |
| | 5 dB | −0.0021 |
| CSI aging (Jakes, 60 km/h) | stale by $\sim$10 ms | −0.0025 (then plateaus) |
| Request delay | 1-frame-stale decision | −0.0109 (pessimistic i.i.d. bound) |

$B_{\max} = 0.20$, $\text{BLER}_L = 0.10$ costs the selector 0.00634 F1, the threshold rule 0.00588 and Fixed $L$ 0.00734 (results/sensitivity/r23_object_message_bler.csv). The selector's margin over Fixed $L$ therefore widens as the cheap link degrades, while its gap to the threshold rule widens against it, from −0.00010 to −0.00056 — the threshold rule leans harder on the feature action ($\rho_F = 0.20$ against 0.12) and is correspondingly less exposed to an unreliable $L$. We report this rather than the favourable half alone; it remains inside the pre-registered margin of 0.005 at every $\text{BLER}_L$ evaluated.

## VII. Conclusion

We presented CA-TOSG, a channel-aware task-oriented semantic granularity selector for V2V cooperative perception. The selector takes per-frame LiDAR-derived perception cues together with an estimated SNR and a channel-type indicator and outputs a per-frame communication mode from the deployed action set $\mathcal{S} = \{E, L, F\}$—ego-only operation $E$ with no request at all, a compact object-level message $L$, or a compressed feature-level message $F$ under 16-QAM coding (the dominated 256-QAM mode is defined but excluded from deployment, Section III-B). The selector is implemented as a lightweight Random Forest that adds no learnable parameters to the perception pipeline; the contribution is not the Random Forest itself but the channel-aware granularity policy, of which the forest is one interpretable implementation (a hand-tuned SNR-threshold rule captures much of the same effect, Section VI-J). It runs in $52.1 \pm 5.6$ ms (P95 = 58.3 ms) per frame on a single CPU core—the slowest of the three frozen selectors, since the deployment claim must hold for all of them—within the 100 ms budget of a 10 Hz LiDAR cycle (Section VI-O).

The results support three conclusions. First, under the evaluated LDPC + QAM feature-transmission setting, fixed feature-level communication is often dominated by compact object-level communication because of channel-induced cliff effects. Second, channel-aware semantic granularity selection requests feature-level communication only when the channel and task state justify it, and its realised AP is reported descriptively rather than as an advantage: the AP a granularity policy can contest is bounded by each split's headroom between the perfect-channel feature ceiling and fixed object-level communication (0.0550 validate, 0.0240 test, 0.0970 Culver-City), and the frozen selector converts at most about a fifth of it (up to 21.3%, and 0.0% on Culver-City at the tightest budget). What transfers to the scene-disjoint test split and the Culver-City domain shift under a frozen selector is therefore the communication saving: 3.7–21.4% of the channel use of fixed feature-level transmission across the three budgets on test. Third, the dominant decision signal is the channel state: a simple SNR-threshold rule tracks the learned selector closely on the channel-averaged payload–F1 frontier and in fact attains a marginally higher realised F1 at every budget, at 1.47–1.97× the channel use; what the pre-registered comparison establishes is non-inferiority within a 0.005 margin at a 34.8% payload reduction against an over-budget nominal threshold — 26.6% against a budget-matched one — not dominance. The granularity policy's gain over object-level communication is itself frame-selective, reaching +0.0660 F1 (95% CI $[+0.0591, +0.0730]$) on the hardest tercile of test frames under a reliable channel; we attribute the limited marginal value of the perception cues to the sharp LDPC + QAM SNR cliff, which makes channel state a near-sufficient statistic. The same boundary appears across detectors: repeated on a second backbone under an equal-budget controlled protocol, the procedure is in-sample effective but does not transfer, with the ego-only action collapsing further than on the mainline (Appendix B). When the feature branch instead uses graceful importance-map JSCC, the estimated SNR becomes uninformative and the picture reverses: the best SNR threshold collapses to object-level performance, and, in-distribution, the cue-based selector beats it using the task cues alone (Appendix A); a selector trained on one split and frozen does not yet transfer this graceful-channel gain across domains, which we leave to future work. Whether task-oriented perception cues are needed may thus be determined by the feature codec's channel response: under the cliff codec evaluated here their contribution is to the shape of the policy rather than to peak accuracy, whereas the graceful-codec half of that statement rests on exploratory evidence gathered under the prior protocol and reported in Appendix A, not re-evaluated under the frozen one.

## Appendix A
## When Are Perception Cues Necessary? LDPC Cliff versus JSCC Graceful Degradation

Protocol note. A prior-protocol arm, not re-evaluated under the frozen P2 protocol: its selector is trained and evaluated inside the codec-comparison pipeline, sharing the retired engine's BLER lookup and 200-realisation CSI convention. Its numbers may not appear in the same sentence as a frozen-replay number, and no main-text claim rests on them.

Under LDPC + QAM a one-line threshold tracks the learned selector (Section VI-J); does that equivalence survive a feature codec without a sharp SNR cliff? We replace the feature branch with importance-map JSCC [6] and re-run the per-frame pipeline: per-frame JSCC F1 is re-cached at each SNR (1,000 frames per SNR, AWGN and Rayleigh), the $\{L, \text{Feature-JSCC}\}$ oracle relabelled and the selector retrained—no perception-backbone change.

The mechanism is a codec property. The LDPC + QAM feature F1 cliffs over the 8–12 dB decoding transition (onset 8.0 dB), so SNR is highly informative; the learned JSCC feature F1 is essentially flat at $\approx 0.89$ across 0–20 dB, so the estimated SNR says almost nothing about whether a feature message will help on a given frame.

This flips the comparison (Table XII). Both codecs are evaluated on the same 2,170 test frames under the identical 200-seed protocol, separating (a) whether the graceful-channel gain exists and resides in the cues—estimated leakage-free in-distribution by 5-fold cross-validation, every frame scored by a selector that never saw it and $\tau$ tuned on the training folds only—from (b) whether a deployed, validate-frozen selector realises it, under the headline cross-split protocol.

(a) In-distribution. Under LDPC + QAM the AWGN edge over the best SNR threshold is only +0.002 F1 (95% CI $[+0.001, +0.003]$): small, and—as the feature ablation shows—not attributable to the perception cues alone (Section VI-E).[7] Under JSCC the best SNR threshold collapses to the Fixed-$L$ operating point—it cannot gate on an uninformative SNR—and the cue-based edge, measured by the in-distribution $k$-fold estimator, widens to +0.022 F1 ($[+0.020, +0.025]$) under AWGN, +0.018 ($[+0.016, +0.020]$) under Rayleigh, and +0.020 ($[+0.018, +0.022]$) under OFDM on the test split, and to +0.012–+0.015 on the Culver-City domain shift (all CIs exclude zero). In the separate 200-realisation held-out comparison on validate frames, the cue-based selector recovers 56–62% of the clairvoyant oracle's headroom over Fixed $L$ across the three channels. The two quantities are distinct and are reported separately: under AWGN the oracle headroom is 0.0291 F1 and the selector recovers +0.0181 of it, with 0.0275 and +0.0158 under Rayleigh and 0.0281 and +0.0158 under OFDM. In-distribution, then, the graceful-channel decision is content-bound and carried by the ego-side cues—a gain no SNR threshold reaches on these frames, and one that (b) below shows does not survive the cross-split move.

(b) Cross-split deployment. The signal does not transfer: deployed on test the frozen selector over-selects the feature action (JSCC C-request rate 0.14 → 0.42) and falls just below Fixed-$L$—edge −0.004 (AWGN), −0.008 (Rayleigh), −0.005 (OFDM) on test, ≈0 on Culver-City. The gain is present but not portable; capturing it under cross-domain deployment, e.g. by online or self-supervised adaptation [26], is future work.

TABLE XII
Learned-selector edge over the best SNR-threshold rule on the same 2,170 test frames (200-seed protocol), reported two ways. In-dist. edge: leakage-free 5-fold cross-validation on the evaluation split (each frame scored by a selector that never saw it, $\tau$ tuned on the training folds only). Deployed edge: selector trained on validate and frozen, applied cross-split (the headline protocol). In-distribution the JSCC edge is 9–11× the LDPC-cliff edge—an SNR threshold cannot reach it—while under LDPC the margin is the selector's function form, not a cue gain (cues add no significant F1, Section VI-E). The deployed JSCC edge is negative: the validate-trained selector does not transfer the graceful-channel gain across the domain shift (Appendix A(b)).

| Feature codec | Channel | In-dist. edge | Deployed edge |
|---|---|---|---|
| LDPC + QAM (cliff) | AWGN | +0.002 $[+0.001, +0.003]$ | −0.002 $[-0.003, -0.001]$ |
| Importance-map JSCC | AWGN | **+0.022** $[+0.020, +0.025]$ | −0.004 $[-0.007, -0.001]$ |
| Importance-map JSCC | Rayleigh | **+0.018** $[+0.016, +0.020]$ | −0.008 $[-0.011, -0.006]$ |
| Importance-map JSCC | OFDM | **+0.020** $[+0.018, +0.022]$ | −0.005 $[-0.008, -0.003]$ |

Whether task-oriented cues are needed is determined by the feature codec's channel response. Under a cliff codec the decision is channel-bound and SNR suffices; under a graceful codec it becomes content-bound and the ego-side cues—which predict whether object-level detection is already adequate—become the operative signal, in-distribution. Closing the cross-split transfer gap (b) is the remaining challenge.

## Appendix B
## Boundary of the Method: a Second Detection Backbone

Convention. This arm was trained and evaluated on the full-collaborator caches, i.e. before the corrigendum of Change-log P0: it is internally consistent (its own budget, its own $N_{cw}$, its own references) and its relative conclusion stands, but its absolute F1 and payload figures are on a different accounting basis from the single-collaborator main experiment and are not tabulated beside it.

Positioning. This appendix reports a limitation, not a generality result. It is placed here because its conclusion is negative: the selection procedure developed in the main text is in-sample effective on a second backbone but does not transfer to held-out splits under the equal-budget protocol. No claim in the main text depends on it.

Every result above uses one perception backbone. To test whether the selection procedure—not the particular features—is what carries the behaviour, we repeat the whole pipeline on a second detector, SECOND [27], under an equal-budget controlled comparison: the feature message is charged the same per-frame channel budget as the mainline ($B_F \equiv 0.99$ Msym, the same $N_{\mathrm{cw}}$ and the same BLER tables), so the two backbones are compared at equal channel cost. We do not claim that SECOND's feature tensor compresses to that size, and we declare no

[7] This in-distribution per-frame edge (+0.002) is distinct from two other LDPC-regime quantities: the feature-ablation gain of the cues (Full vs. channel-state-only, +0.0090, 95% CI $[+0.0089, +0.0091]$, Section VI-E) and the matched-payload margin over the retuned threshold (+0.0032, Section VI-A). The three measure the selector along the cue, the equal-bandwidth, and the per-frame axis respectively. Under the corrected single-collaborator convention all three are positive, with the ordering $+0.002 < +0.0032 < +0.0090$: the cue axis, smallest and indistinguishable from zero under the retired full-collaborator accounting (−0.0002), is now the largest of the three — weakening the single message widened the gap the cues have to close, and they close part of it.

TABLE XIII
Second backbone (SECOND), equal-budget controlled. Paired ΔF1 of the frozen selector against the SNR-threshold rule tuned to the same budget, over the same 200 CSI samplings; 95% frame-level paired bootstrap intervals. Positive favours CA-TOSG. Second-backbone arm, not deployed; descriptive, no decision is taken from this table.

| Split | $B_{\max} = 0.10$ | $B_{\max} = 0.20$ | $B_{\max} = 0.30$ |
|---|---|---|---|
| Validate (in-sample) | +0.0058 [+0.0057, +0.0059] | +0.0095 [+0.0094, +0.0096] | +0.0081 [+0.0081, +0.0082] |
| OPV2V test | −0.0014 [−0.0015, −0.0013] | −0.0047 [−0.0048, −0.0046] | −0.0066 [−0.0067, −0.0065] |
| Culver-City | −0.0029 [−0.0030, −0.0028] | −0.0117 [−0.0119, −0.0115] | −0.0167 [−0.0169, −0.0165] |

codec for it.[8]

Both SECOND checkpoints are the official OpenCOOD ones, and we verify them before use rather than trusting the file: after a lossless axis-order conversion required by the spconv version in our environment, official inference reproduces the published AP@0.7 to +0.0002 on both splits for the late-fusion model (0.7752 vs. 0.775; 0.6822 vs. 0.682) and to within +0.0019 for the intermediate model. The three per-frame branches ($E$, $L$, $F$) are then re-derived with the same scorer, the same canonical union ground truth and the same IoU-0.5 convention as the mainline, and the entire downstream pipeline—grid expansion, scene-level 9-fold LOSO, the budget walk, and the 200-realisation paired replay—is re-run with the mainline code, each stage first verified to reproduce its own committed mainline product bit-for-bit.

The result is that CA-TOSG is in-sample effective but does not transfer under the equal-budget protocol. Table XIII gives the paired comparison against an SNR-threshold rule tuned on validate for the same target budget. On validate—the split the selector is trained on—it is ahead at every budget (+0.0058 to +0.0095 F1). On both held-out splits it is behind at every budget, with all confidence intervals excluding zero.

Two mechanisms account for the gap, and both are visible in the arm's own numbers. First, the $L$–$F$ margin is narrower on this backbone: SECOND's object-level branch scores 0.869 / 0.879 / 0.879 mean per-frame F1 on validate / test / Culver-City against a compressed-feature branch at 0.897 / 0.904 / 0.939, so there is simply less to gain from choosing correctly, and the same selection error costs proportionally more. Second, the budget walk selects a different candidate family: all three frozen SECOND selectors carry class_weight=balanced, whereas all three mainline selectors carry class_weight=None. The reweighted family fits the in-sample class mix well—agreement with the oracle is 0.833 on validate—but transfers poorly, falling to 0.578 on test and 0.534 on Culver-City, where the selector degenerates to near-always-$L$ ($E$ recall 0.027 on test and 0.000 on Culver-City). The $E$-collapse of Section VI-H is therefore worse on this backbone, not milder: under Rayleigh the selector's $\rho_E$ is 0.002 (test) and 0.000 (Culver-City) while the oracle spends 0.157 and 0.133. The large payload reductions this arm shows (67–90%) follow from that collapse to $L$ and are not evidence of good selection.

We report this as a boundary of the present method rather than a tuning target: nothing was retrained and no threshold was adjusted after seeing these numbers.

[8] The transmitted tensor was measured, not assumed: SECOND's cross-link BEV tensor is 6,758,400 elements per collaborator before compression and 352,000 at the auto-encoder bottleneck. Both are recorded as measurements; neither is used as the operative payload under the equal-budget convention.

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